





TOOL WEAR

preliminary to speak about machining economics
(optimization in the process parameter choice)

Tool Life and Machining Economics

↓
optimization of cutting condition

Groover M.P.: Chapters 19 and 20



The student:

- understands the mechanism of cutting tool wear;
- knows materials for cutting tools;
- knows about machining economics.





THREE MODES OF TOOL FAILURE

when machining, we use tools
subjected to failures
(possible kind of fracture)

3

POSSIBLE FAILURES:

- 1. **Fracture failure** if too high load, fracture of insert
 - **Cutting forces** and/or dynamic vibrations become excessive, leading to brittle fracture

↑ to avoid! WORST CASES
↓
- 2. **Temperature failure** increase of temp. above the supported limit
 - **Cutting temperature** is too high for the tool material

failure because in interface of removal material, too high temperature respect tool resistance → FAILS of tool
- 3. **Gradual wear**
 - Gradual wearing of the cutting tool ← it can be predicted, use information to set up machining and optimize cutting conditions

↑ increase of wear during time
to set-up machining and optimize cutting



Behind tool wear!

Phenomena causing tool wear are of the following nature:

- **Mechanical:** mainly due to the abrasion of the chip on the cutting tool rake face;
- **Thermal:** during cutting temperatures rise noticeably;
- **Chemical:** with the temperature increase, we can observe oxidation or solid state diffusion mechanisms that reduce mechanical characteristics of the tool.



Possible mechanisms of wear formation: *phenomenon of wear*

GRADUALLY



- **Abrasion:** the tool meets high hardness particles; *damage the tool gradually*
- **Diffusion:** atoms passage between part and tool materials; *→ due to mechanical phenom... ATOMS from tool goes inside the material → the atoms are exchanged between TOOL ↔ WORKPIECE*
- **Oxidation:** oxygen from the atmosphere forms easily removable oxides; *oxygen react with material... due to temperature also → create abrasion*
- **Adhesion:** phenomenon of the built-up edge; (BUE) *(BUE) to AVOID! modify the tool geometry → break material that remain on surface, RAKE* *↑ abrasion*
- **Plastic deformation:** softening due to mechanical and thermal effects; *change the work condition*
- **Fatigue:** due to mechanical and thermal cycles. *tool used in machining → affected by cycles of mech/thermal loads
→ FATIGUE → FRACTURE*

more CHEMICAL phenomenon

- Diffusion: [tool atoms combined with the WORKPIECE atoms]
small Relevant Impact

Single atom of one of the two material goes on the other material
(atom of chip comes on tool -- and opposite) → due to temperature
+ pressure in contact

IF removing atoms in the tool, it combine
(with others → HARD + BRITTLE (easy separation)) → creating abrasion
small relevant impact

- Abrasion due to the sliding between tool and worked material
hard element inside workpiece material damaging GRADUALLY the tool
on both workpiece and tool

- contact between materials, wear on the FLANK
- during chip formation, it flows on RAKE face...
secondary shear stress due to flowing chip → wear phenomenon
⇒ ABRASION

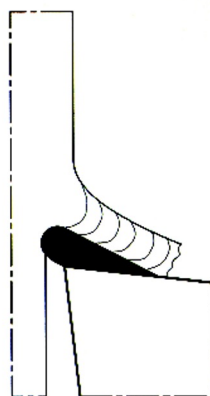
- Oxidation: due to Temperature, oxigene React with material of tool,
create OXIDE (Brittle/hard material) → removed due to stress remaining
between tool / chip = abrasion

- LOAD + TEMPERATURE ↑ ⇒ PLASTIC DEF high T change condition of
the material

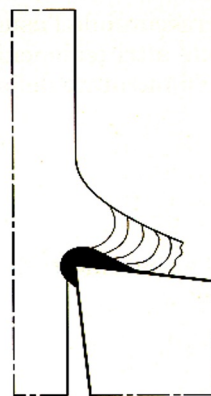
FATIGUE periodical cycle of mech, term. loads → creating fatigue
on the tool
↓
Ready to FRACTURE



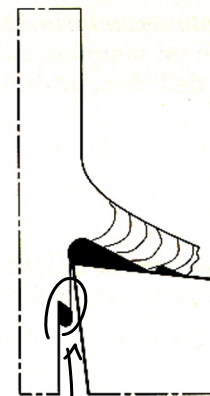
Built-up edge



a)

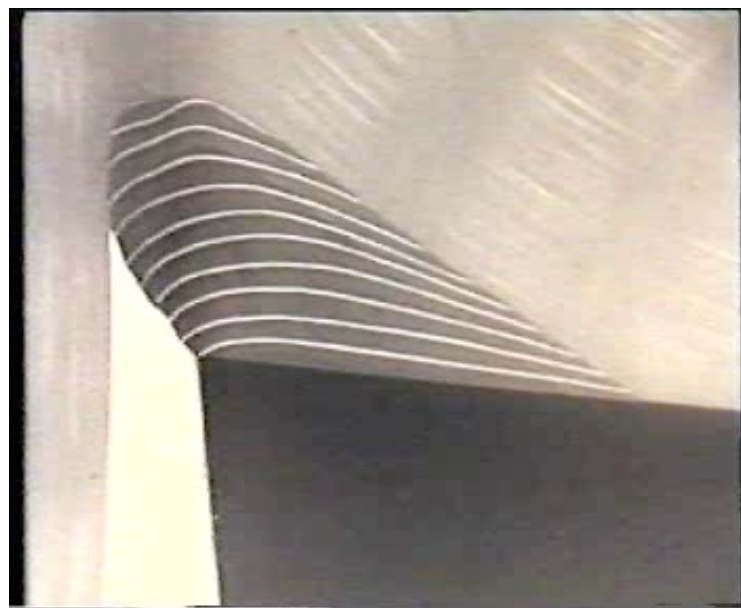


b)



c)

material removing on surface
(Abrasion)
↑



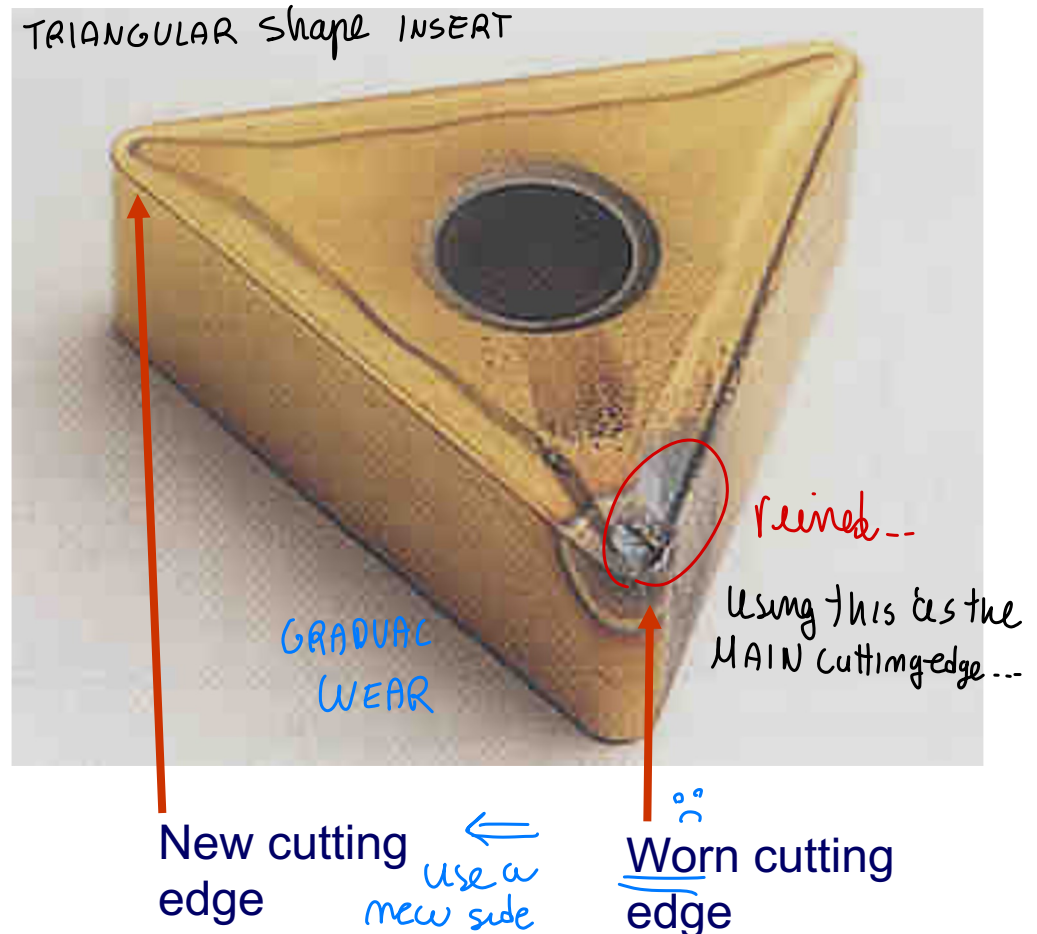
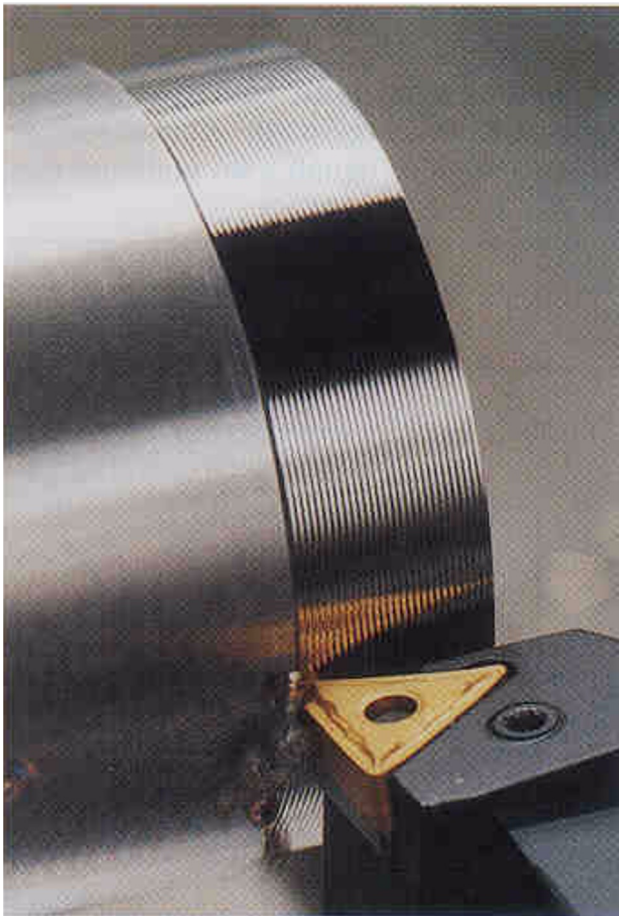
TOOL WEAR

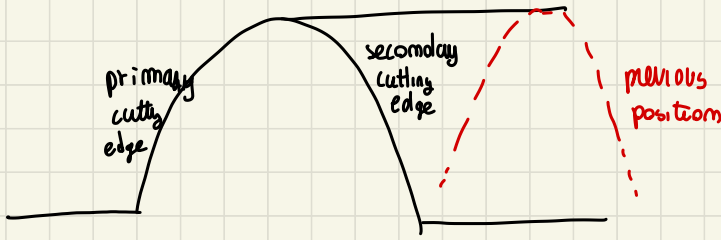
○ Relevant factors influencing the tool wear:

- Tool characteristics;
- Temperature of the cutting zone (cutting speed); ↗ related to V_c
- Cutting parameters; (i.e. cutting speed / material thickness a_p), feed, depth of cut--)
- Cooling of the cutting zone; ← material used to cool down the area
- Presence of thermal cycles;
- Lubrication of the cutting zone;
- Chemical affinity between the tool and the work material; ↑ Diffusion and Abrasion influenced by this--
- ... Chemical aspect ⇒ how easy they can INTERACT



During machining tool **gradual wear** occurs.
Gradual wear is the **preferred mode of tool failure**, because it leads to the longest possible use of the tool.





portion of cutting edge
involved in the cutting

→ we see the
WEAR OF
THE INSERT

Gradual
Wear

that can be PREDICTED/estimated and sometimes control



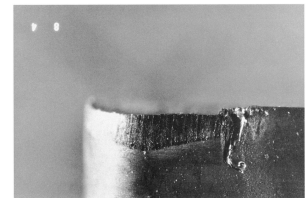
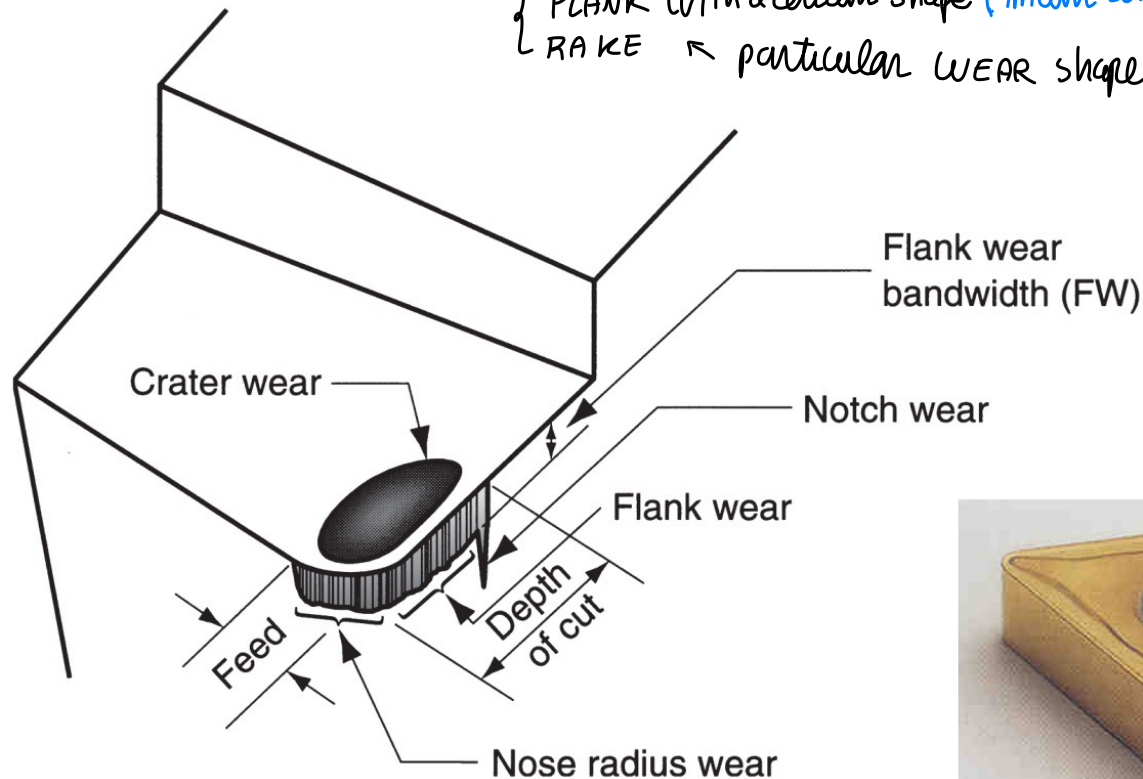
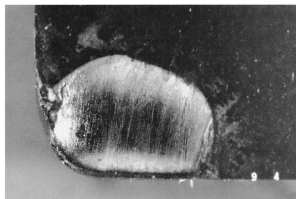
TOOL WEAR

we may have different wear shapes

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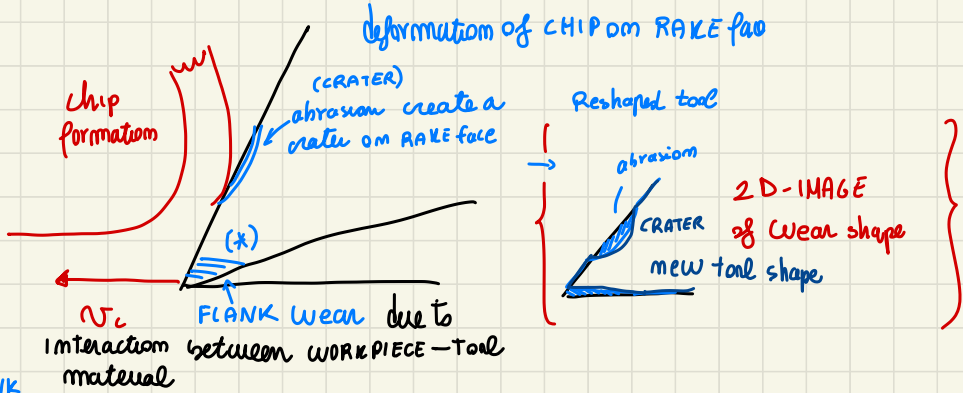
Worn cutting tool, showing the principal locations and types of wear that occur.

Looking wear shapes.. we can have different shapes on different surfaces
FLANK with a certain shape (main cutting edge)
RAKE $\leftarrow$ particular WEAR shape $\Rightarrow$



particular RAKE phase situation (WEAR shape)

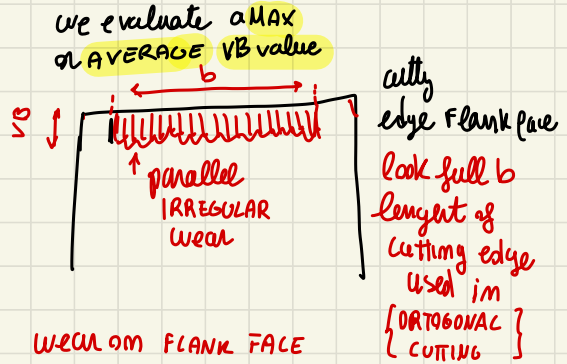
ORTOGONAL CUTTING



{ SIDE VIEW }

measure FLANK WEAR $\rightarrow$ VB

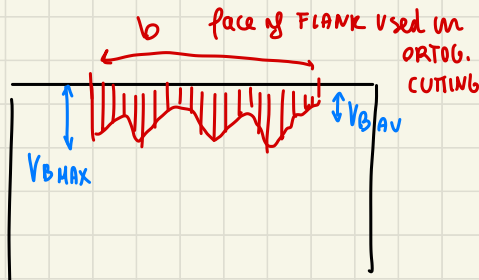
measuring referring to original cutting edge



from an initial perfect cutting edge shape of the FLANK phase ...

Regular wear on that side

flank wear greater on some position boundaries





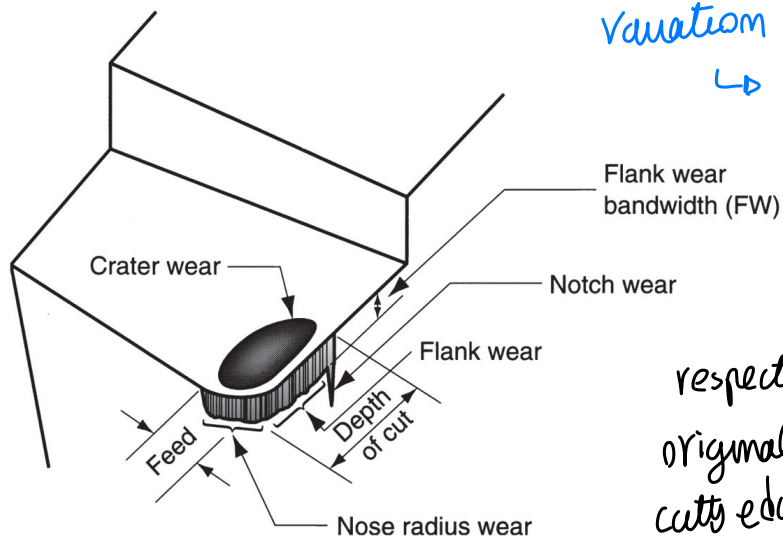
TOOL WEAR

We can measure and monitor ¹⁰ this variation of the tool shape due to WEAR

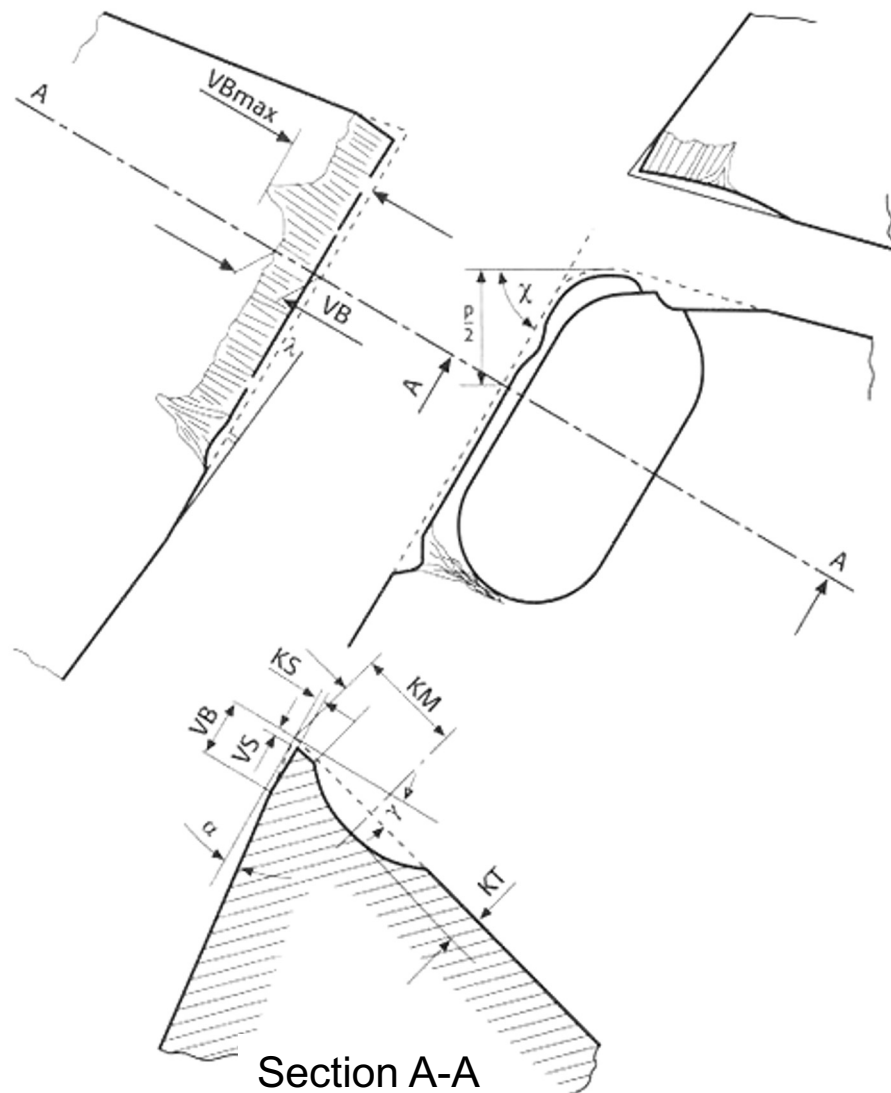
Gradual wear occurs at two locations on a tool:

- **Crater wear**: occurs on top rake face. (due to CHIPS deforming here)
- **Flank wear**: occurs on flank (side of tool). (due to CUTTING CONTACT)

you can measure the tool shape variation

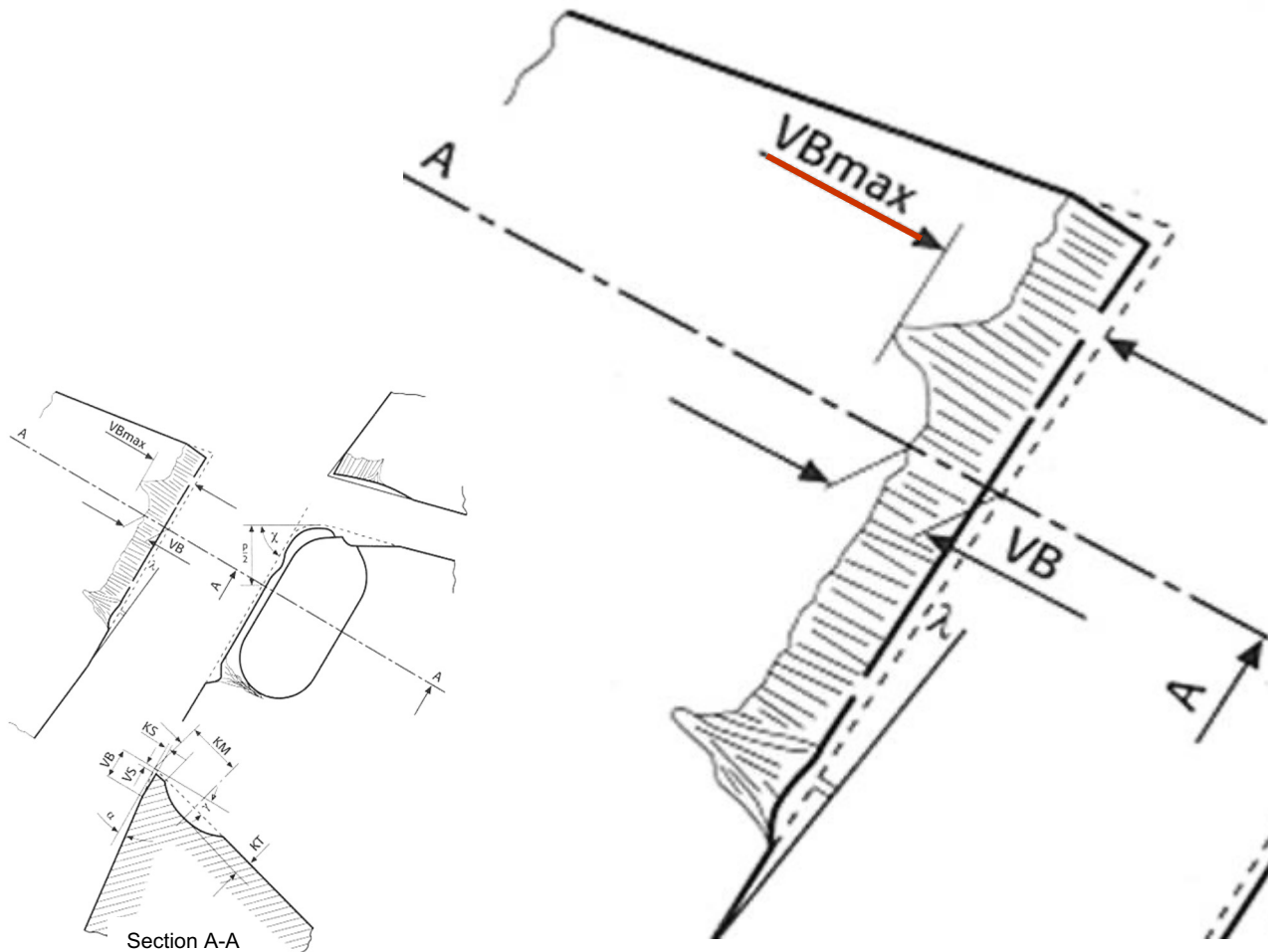


respect original cut's edge





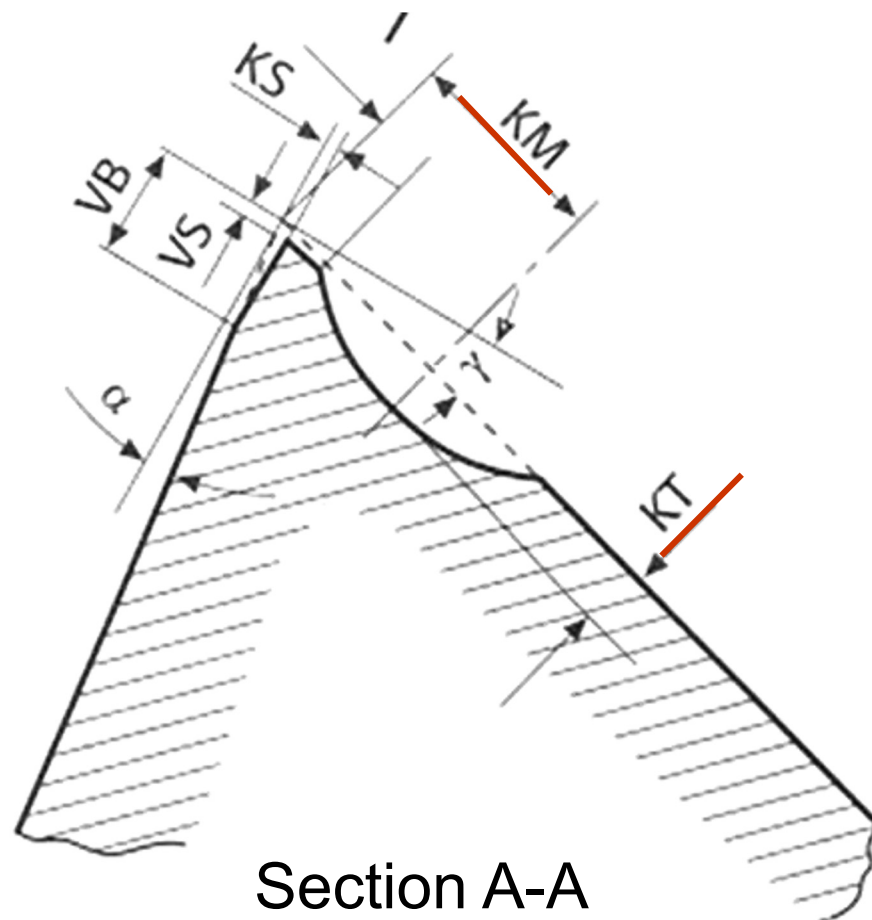
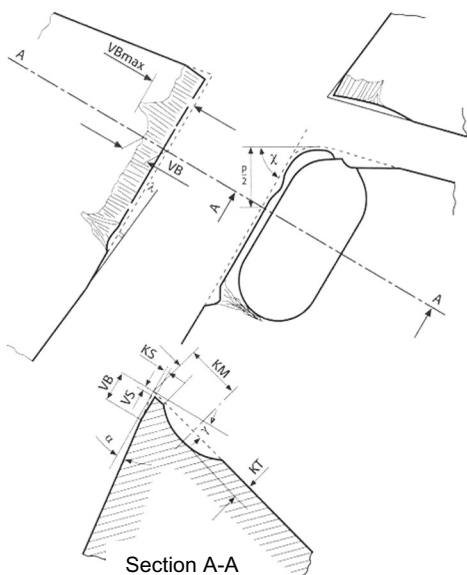
Flank wear affects dimensional accuracy and surface finish. It is quantified mainly by the parameter **VB** (average VB and VB_{max}).



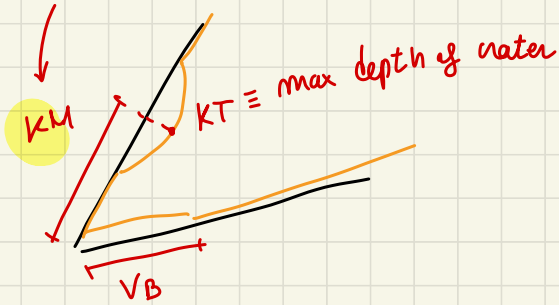


Crater wear weakens the tool. It is quantified by:

- ↑
PARAMETERS
to
characterize
WEAR
- **KT**: crater depth; *We are interested in MAX DEPTH (KT)*
 - **KM**: position of the deepest point of the crater measured from the original cutting edge. *and also interested in the DEPTH MAX position (KM) respect to original depth, measured parallel to RAKE face*



to characterize the CRATER



Lo to characterize the
behaviour of the GRADUAL
TOOL WEAR
Respect
TIME...



TOOL WEAR

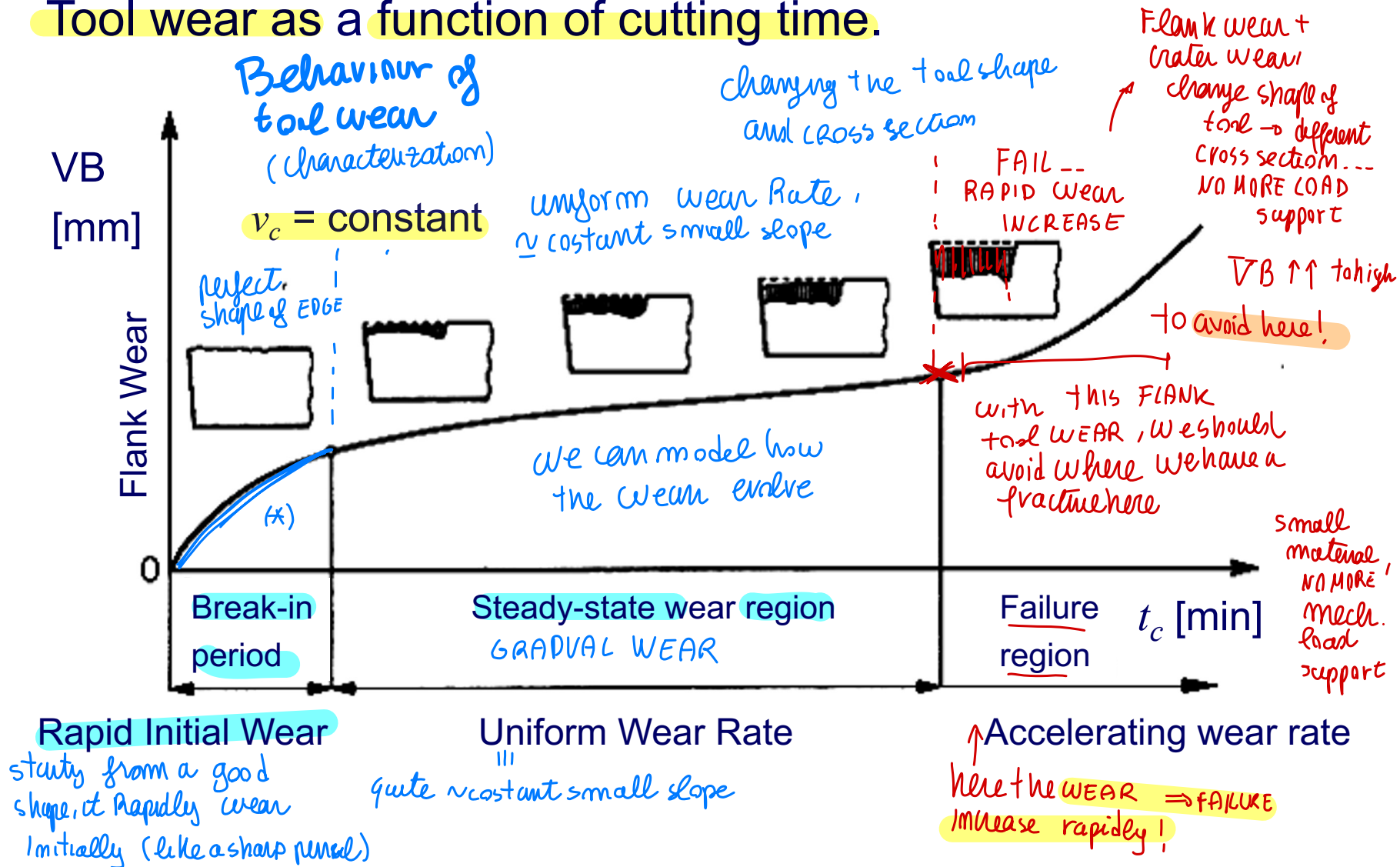
ON TIME

classical shape of how it evolve → to know behavior

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HOW FLANK WEAR VB evolve during CUTTING TIME (t_c) (CONTACT)

Tool wear as a function of cutting time.

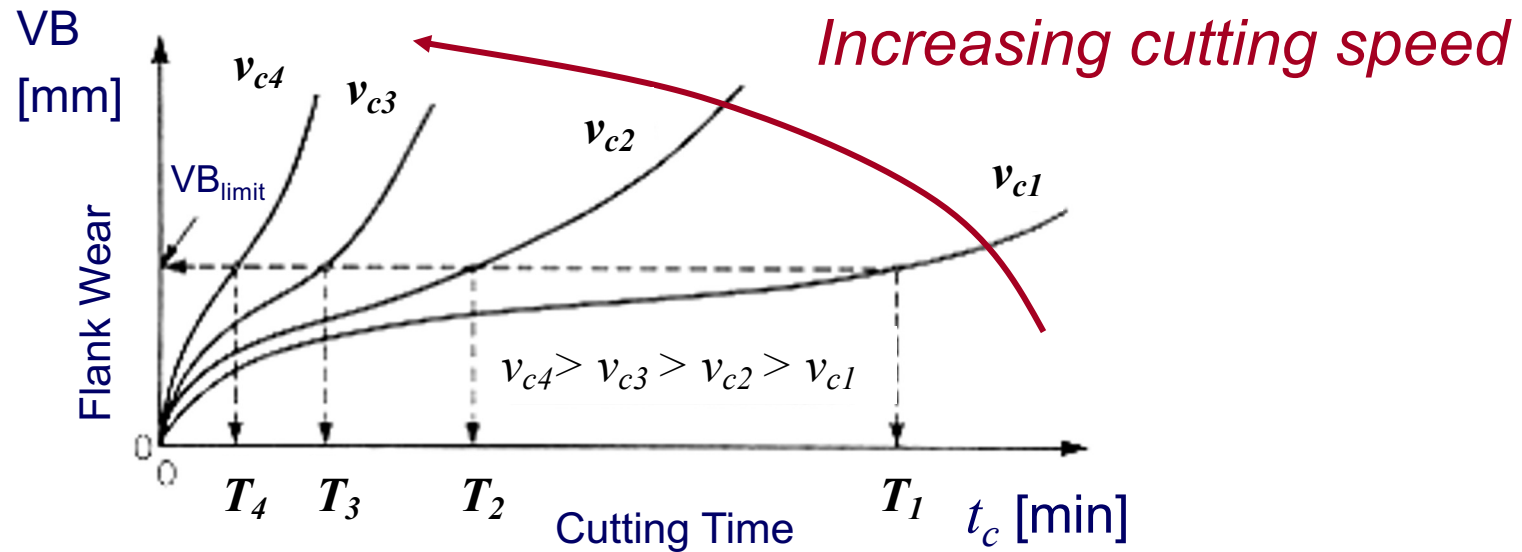




Effect of cutting speed on flank wear (VB)

Behaviour of Flank Wear respect time

the increase of V_c CAUSES
a RAPIDLY WEAR

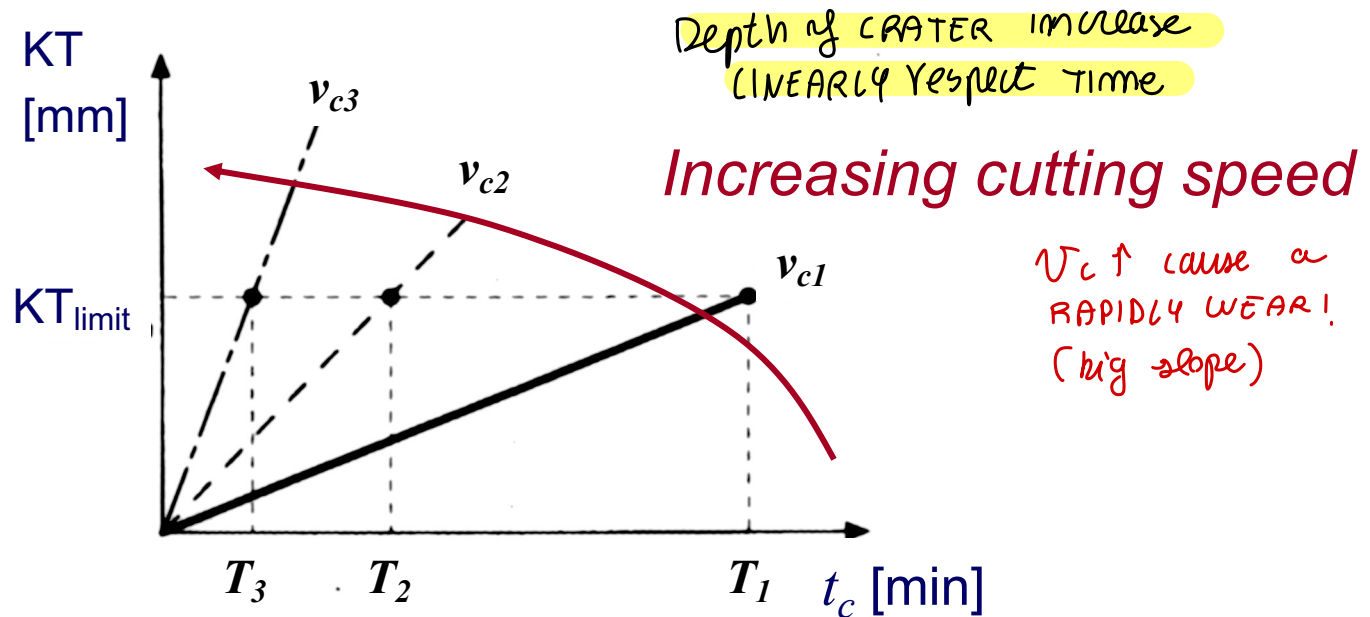


Increasing V_c cutting velocity, you have
a greater WEAR of the piece



Effect of cutting speed on rake wear (KT)

- similarly: the parameter characterizing the crater KT increase more if $v_c \uparrow$

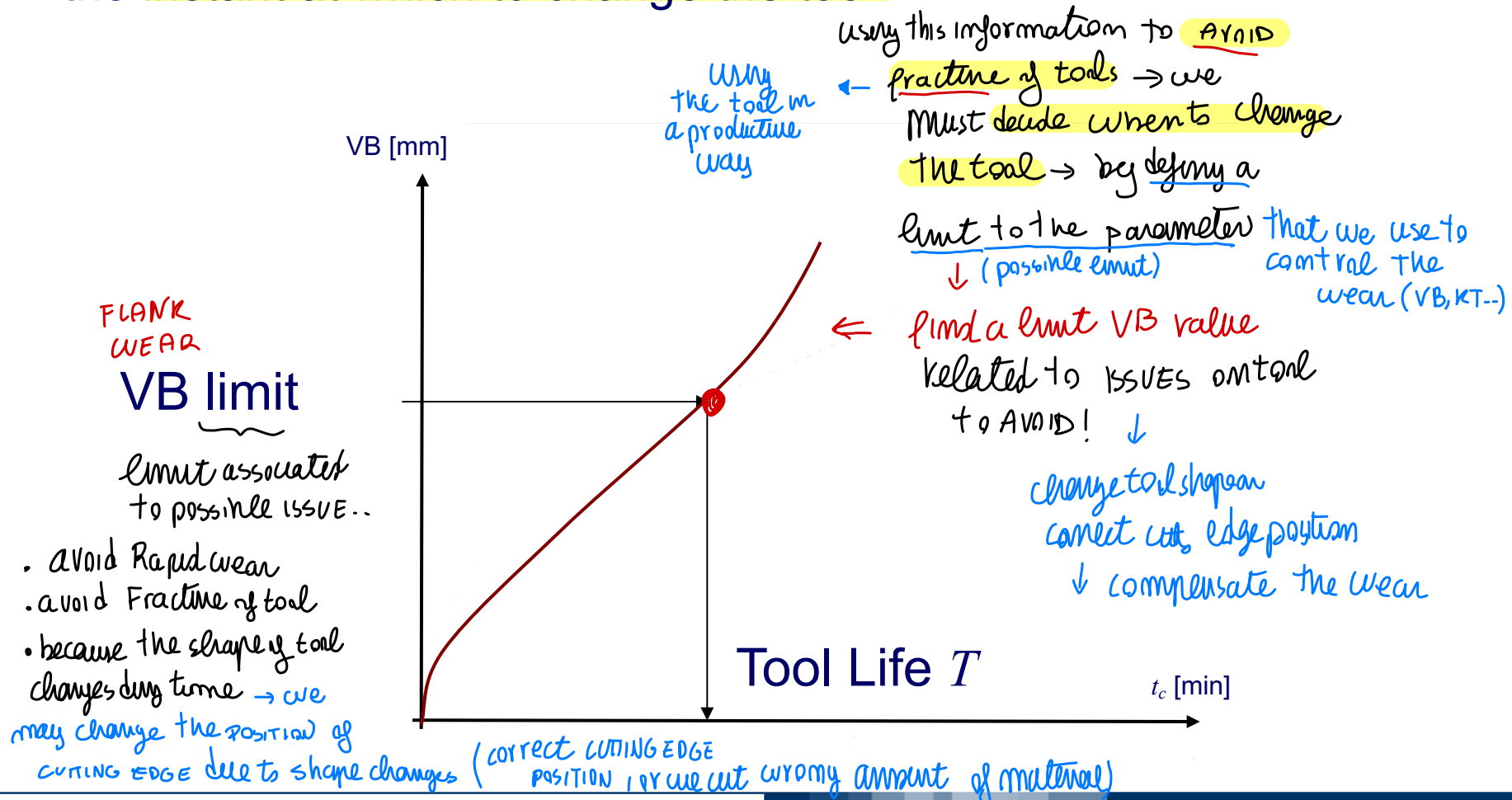


the CRATER
increases ~ linearly, if $v_c \uparrow$ it wears more ---



TOOL LIFE: FAILURE CRITERIA

We must define a **tool failure criterion** in terms of maximum wear level (for example $VB = 0,3 \text{ mm}$, $KT/KM = 0,1$) to determine the instant at which to change the tool.





TOOL LIFE: FAILURE CRITERIA

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When Reached a VALUE of VB limit, enough
BIG... PORTION of tool SLIDING on final surface ---
PROBLEM on final surface

↓ WE MUST
define a
LIMIT!

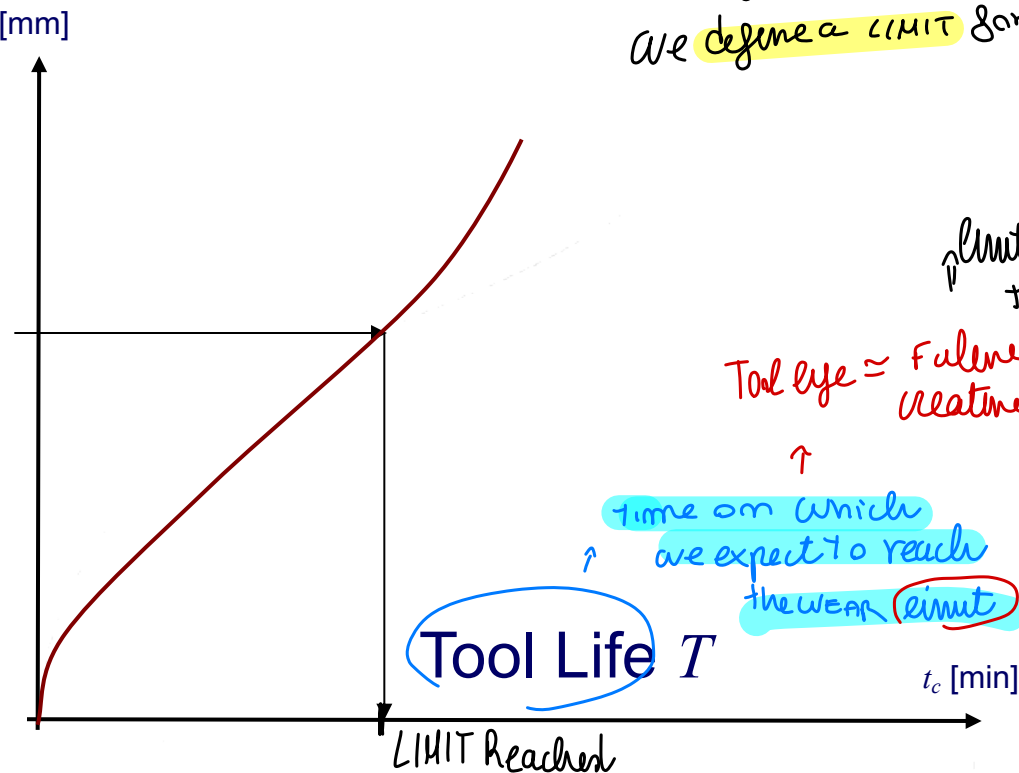
VB limit

Knowing how gradual
change can set-up a LIMIT

↓
We can define the
TOOL LIFE

We can change cutting edge position
to have less wear

↓
We define a LIMIT for any reason



↑ limit of wear
time

Tool life $\approx$ Failure
lifetime

↑
time on which
we expect to reach
the wear limit

↑ define a
Failure criteria

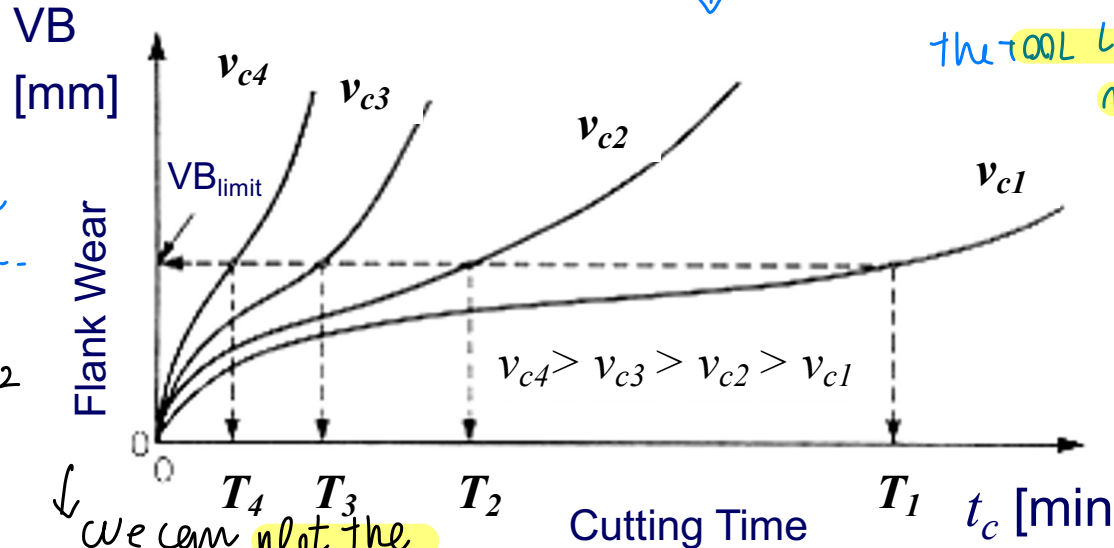


TOOL LIFE CURVES

considering different DIAGRAMS of VB
because a change of v_c changes the WEAR..

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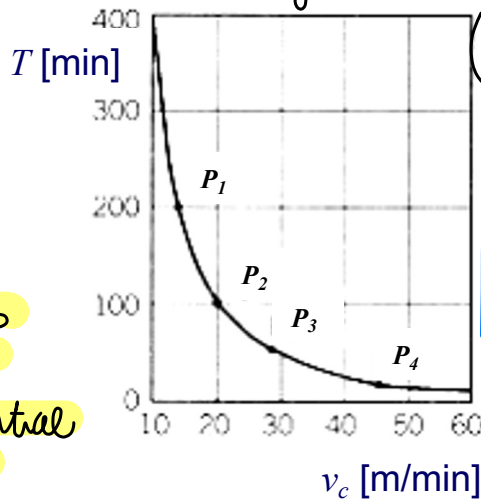
considering v_{c1} we have a certain T_1 related to the VB limit..
and v_{c2} same same a different T_2



the TOOL LIFE decrease as $v_c \uparrow$ (less resistance)

TOOL LIFE related to tool failure criterion
↓
and changes respect v_c

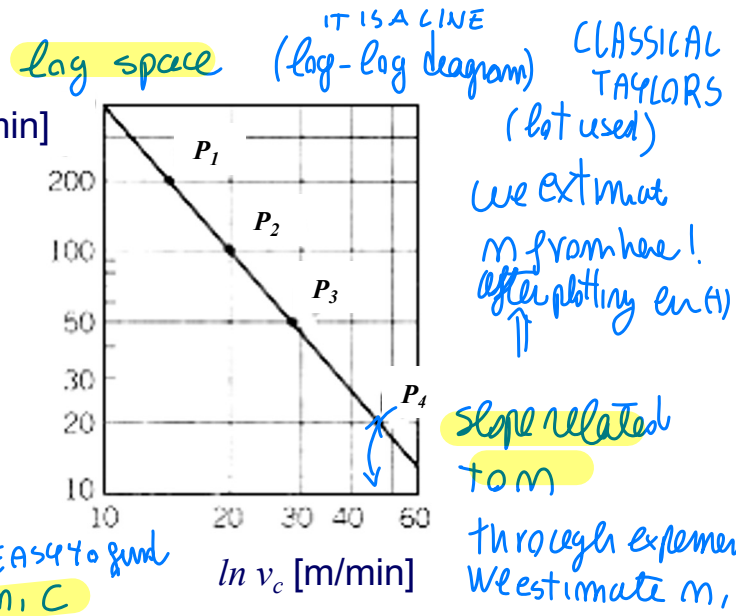
we can plot the Tool life respect v_c



PLOT $T(v_c)$

we see points approx with an exponential decrease of T

(WEAR OF TOOL) $(\ln T [\min])$
Classical Taylor's Law
(empirical law)
next slide



EASIER to find m, C

slope related to m
through experiment we estimate m, C



Classical Taylor's law

Knowing the Behaviour of gradual wear respect time and defined a FAILURE CRITERION (VB limit) $\Rightarrow$ we can evaluate $\forall v_c$ a Tool life value T

empirical law that from the VB (t_c) and $T(v_c)$ diagram

$\leftarrow$ relate cutting speed and tool life given the situation of work

$v_c T^n = C$
situation of work

(given by TOOL MANUFACTURER)

- v_c : cutting speed,
- T : tool duration or tool life,
- n : exponent depending mainly on the cutting tool material,
- C : parameter depending on the workpiece material and on the other machining parameters (feed and cutting depth).

PROCESS

few experiment to have high INFO for LIFE PREDICTION

$\downarrow$ defining a failure criteria, we can evaluate $\forall v_c$ the TOOL LIFE ($T = T(v_c)$)

Generalized Taylor's law

and depth of cut

More generalized equation

MORE COMPLEX and COSTLY to parameter define

$$v_c T^n f^m (a_p^q) = C^*$$

considering also feed

obtained through experiment

more precise expression, similar to the classical one

$$T = T(v_c, f, a_p)$$

function of process parameters

to have the constant of the Taylor's law n, C we need to perform experiment

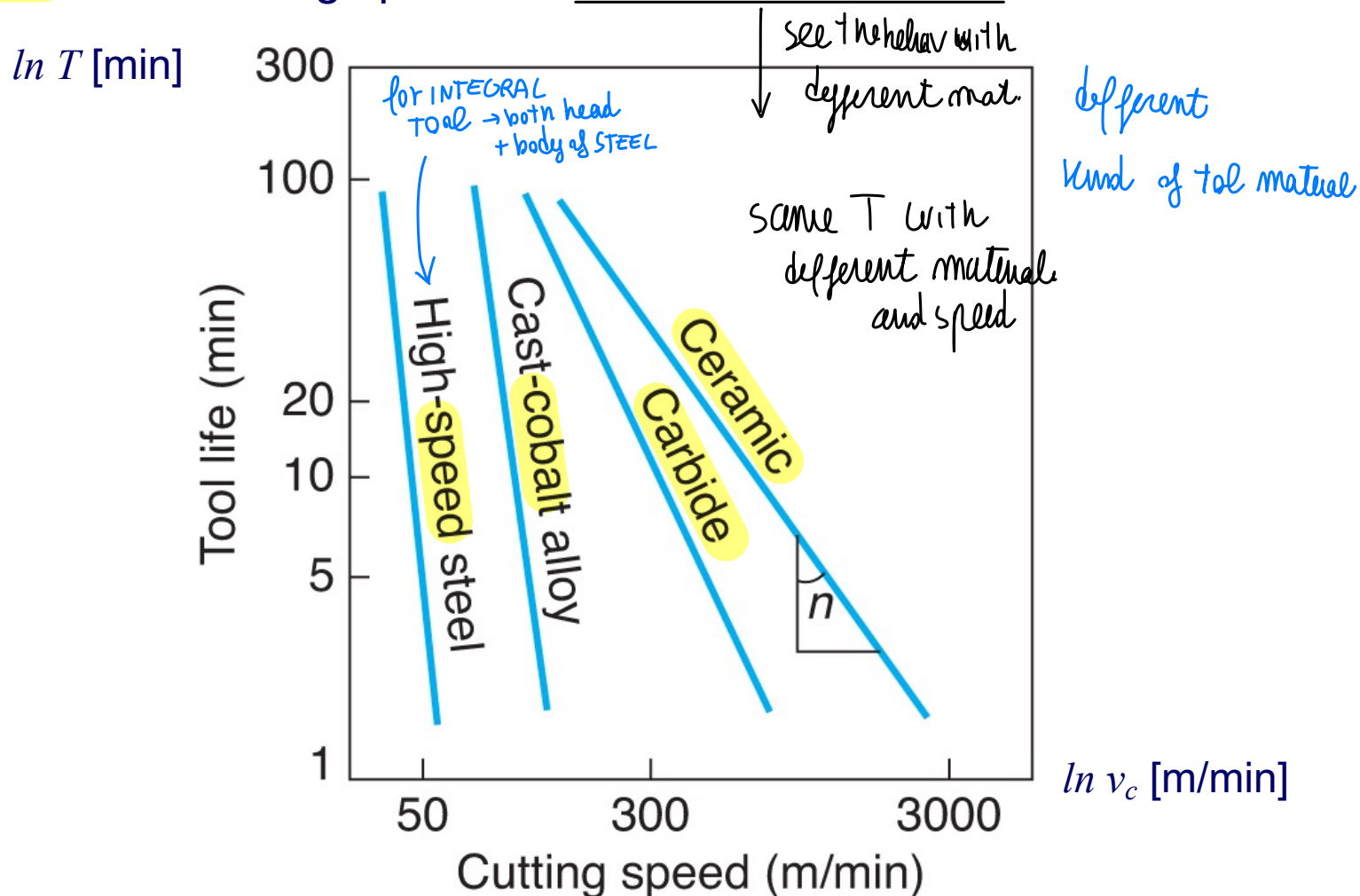


TOOL LIFE LAWS

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examples

Tool life versus cutting speed for different tool materials.





TOOL LIFE CURVES

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Tool life versus cutting speed. *log-log diagram* (NOT simply linear!)

REAL DIAGRAM making many experiment you find that Real diag, NOT AS TAYLOR LAW

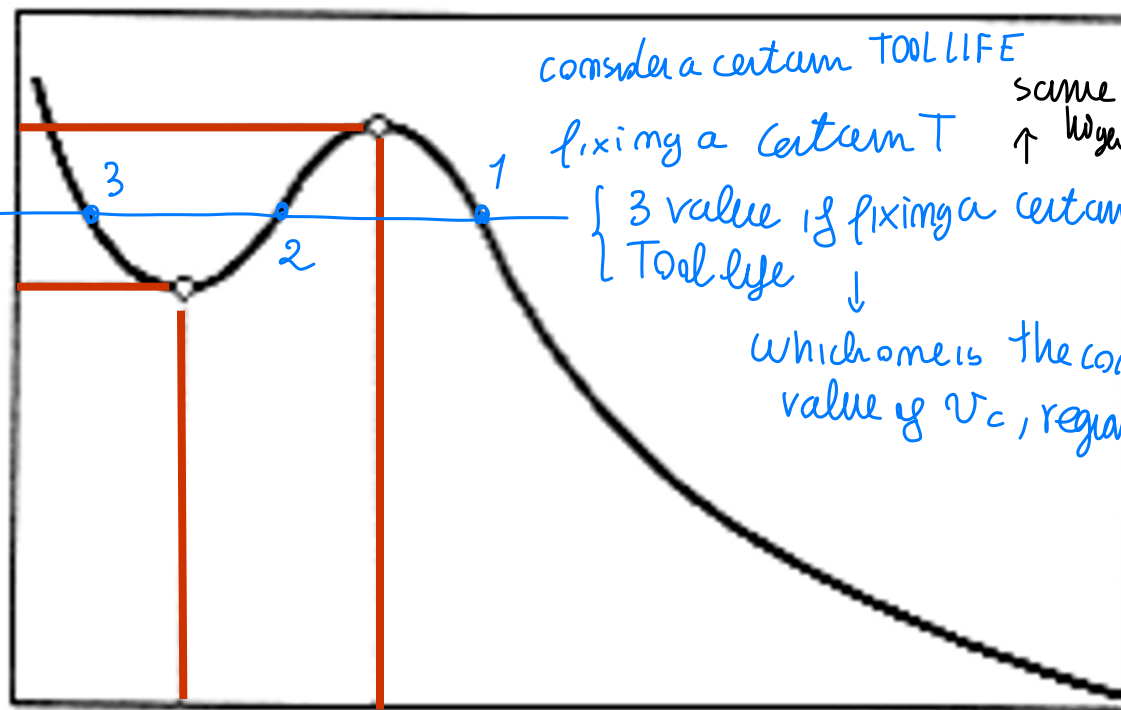
$\ln T$ [min]

$\ln T_2$

T

$\ln T_1$

Tool Life



$\ln v_{c,1}$

$\ln v_{c,2}$

$\ln v_c$ [m/min]

Cutting Speed

even if
Real diagram
is this one --
why we
can use
Taylor's law
with accuracy (CORRECT)

we apply Taylor's law
even if the log-log
diagram is different

consider a certain TOOL LIFE

fixing a certain T

{ 3 value if fixing a certain
Tool life }

↓
which one is the convenient
value of v_c , region to work

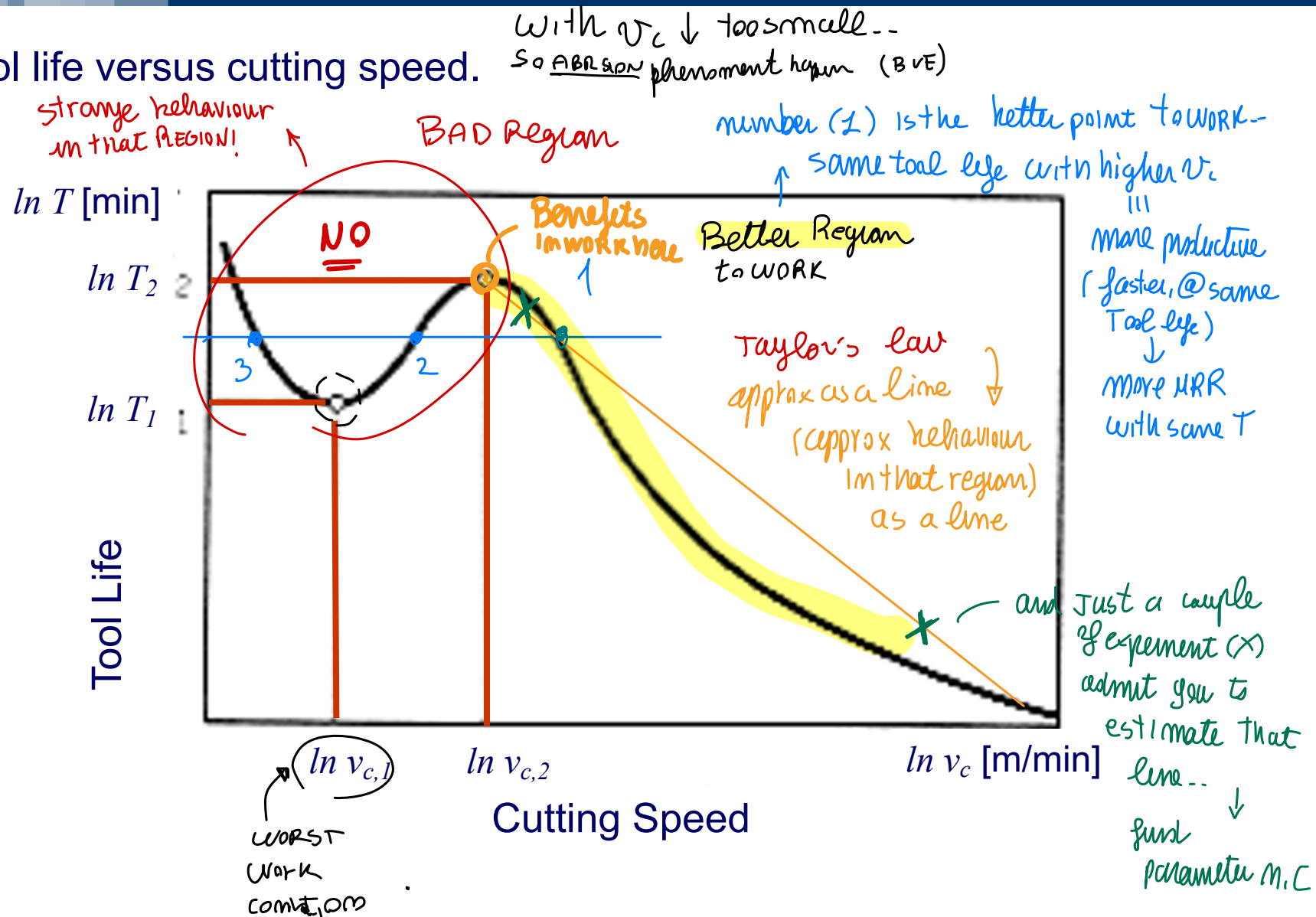
same tool life with
↑ higher cutting speed



TOOL LIFE CURVES

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Tool life versus cutting speed.



We can have same T (tool life) by different V_c faster!

So we approximate the behavior of T as a line

- for point (2), (3) with low V_c value of cutting speed...
you always cut **BUT** cutting with a slow velocity
you cause a Bad phenomenon
when you cut slow $\rightarrow$ adhesion and BUE

We don't
desire too small velocity $\rightarrow$ create condition of BUE undesired

The condition of (V_{c1}) favour BUE creation... local min
WORST condition

FAST tool life reduction @ T_1 respect V_{c1} (fast T reduction)

(1) With same T
but more productive! **Better!**



Main requirements for a cutting tool:

WISH LIST
OF PROPERTIES

to have a
GOOD
tool we requires...

- Toughness
- Wear resistance
- Hardness at high temperature
- Heat conductivity *good temperature behavior
controlling wear --* ← *good controlling of Temp,
it flows well*
- Low thermal expansion coefficient → *does modify small
the shape for high Temp.*
- Low friction coefficient
- Chemical inertia *to avoid diffusion, atmosphere*
- Limited cost *(low cost wanted!)*

↑ we have to
obtain a TRADE-OFF of that req. using well materials



CUTTING TOOL MATERIALS (TOOL FAMILY and helman)

LIST of possible materials (Families of possible tools)

■ Carbon steels *considering different applications*

■ High-speed steels

■ Cast Alloys

■ Carbides (coated or uncoated)

- WC-Co, *different kind of coats*
- WC-TiC-Co,
- WC-TiC-TaC-Co

■ Ceramics

- Al_2O_3 , Si_3N_4

■ Cermets (ceramic+metals)

■ CBN: Cubic Boron Nitride → *material second only to DIAMOND, HARD!*

■ **Diamond: natural, artificial, polycrystalline.** → *common in GRINDING, enabling to obtain accurate surface*
Best, HARDER common cutting material (to obtain accurate surface finish)
good roughness ↑

all this materials are so that we can increase v_c with $T = \text{const}$

+ v_c

(with constant Tool Life)

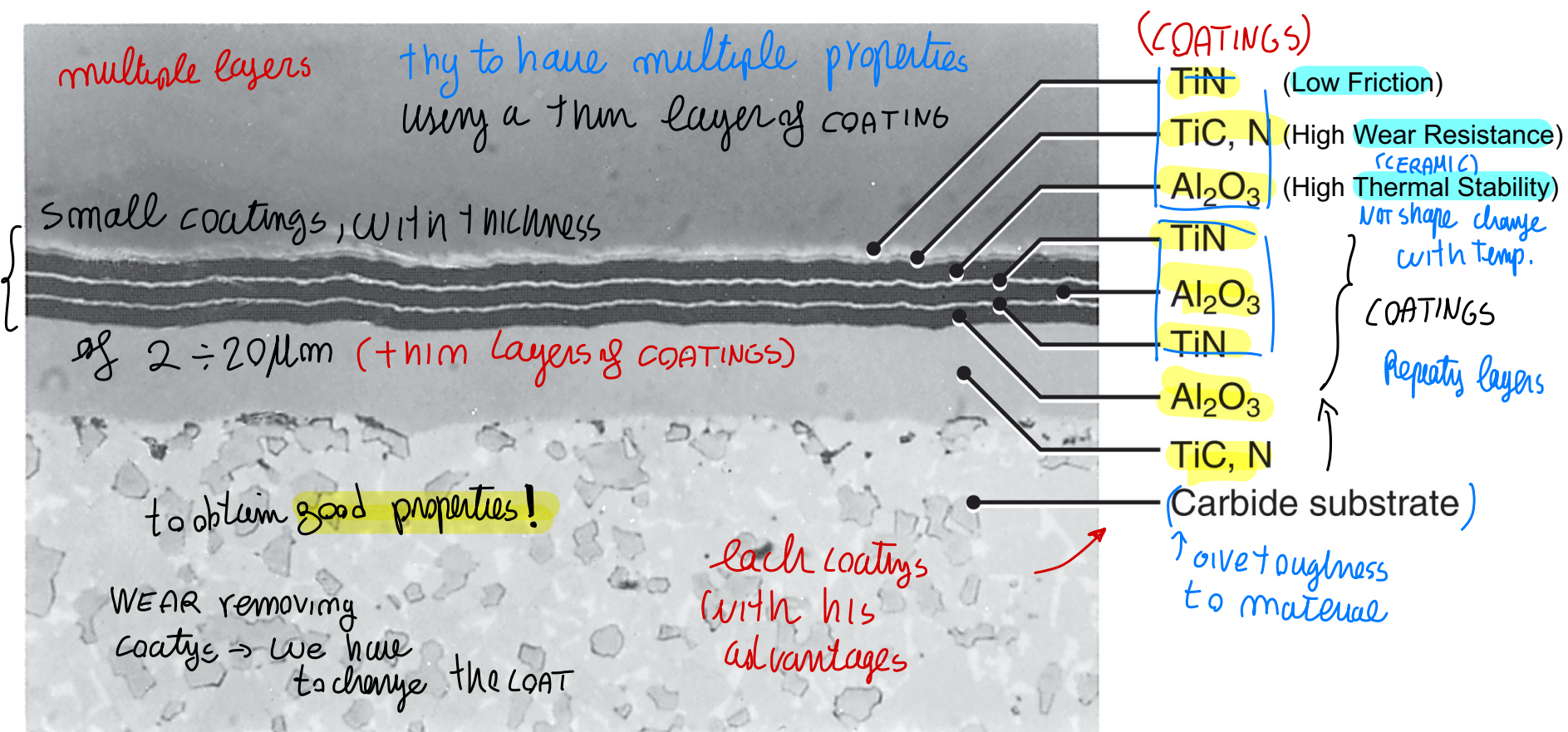
With $T = \text{const}$

you can increase v_c



CUTTING TOOL MATERIALS

Multiphase coatings on a tungsten-carbide substrate; three alternating layers of aluminum oxide are separated by very thin layers of titanium nitride. Inserts with as many as 13 layers of coatings have been made. **Coating** thicknesses are typically in the range from 2 to 20 μm . *combine thin COATING layers*



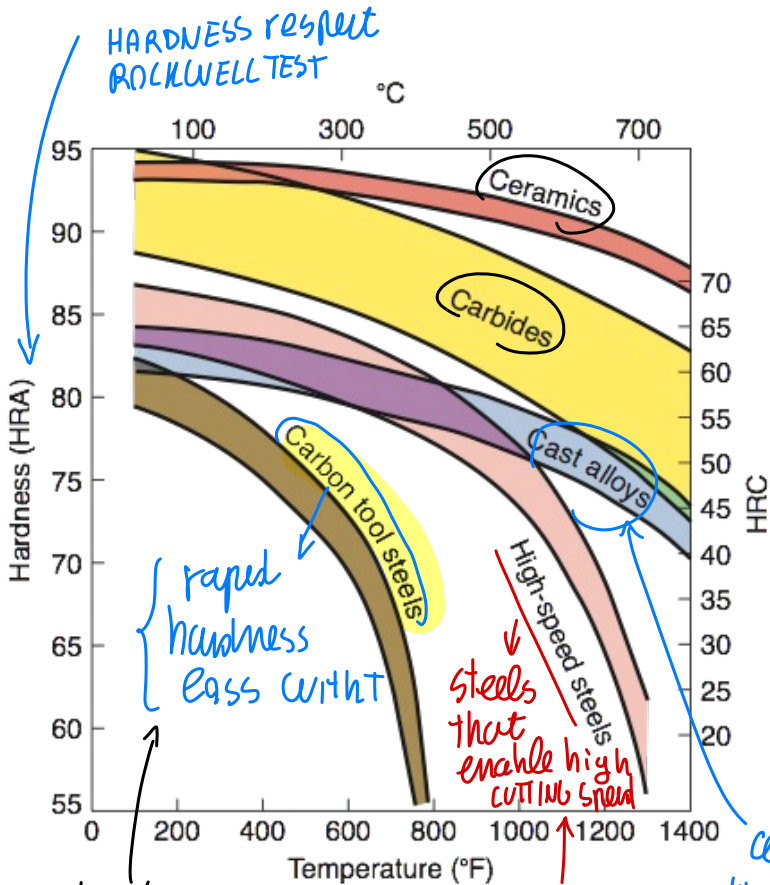


CUTTING TOOL MATERIALS

good understanding of PROPERTIES

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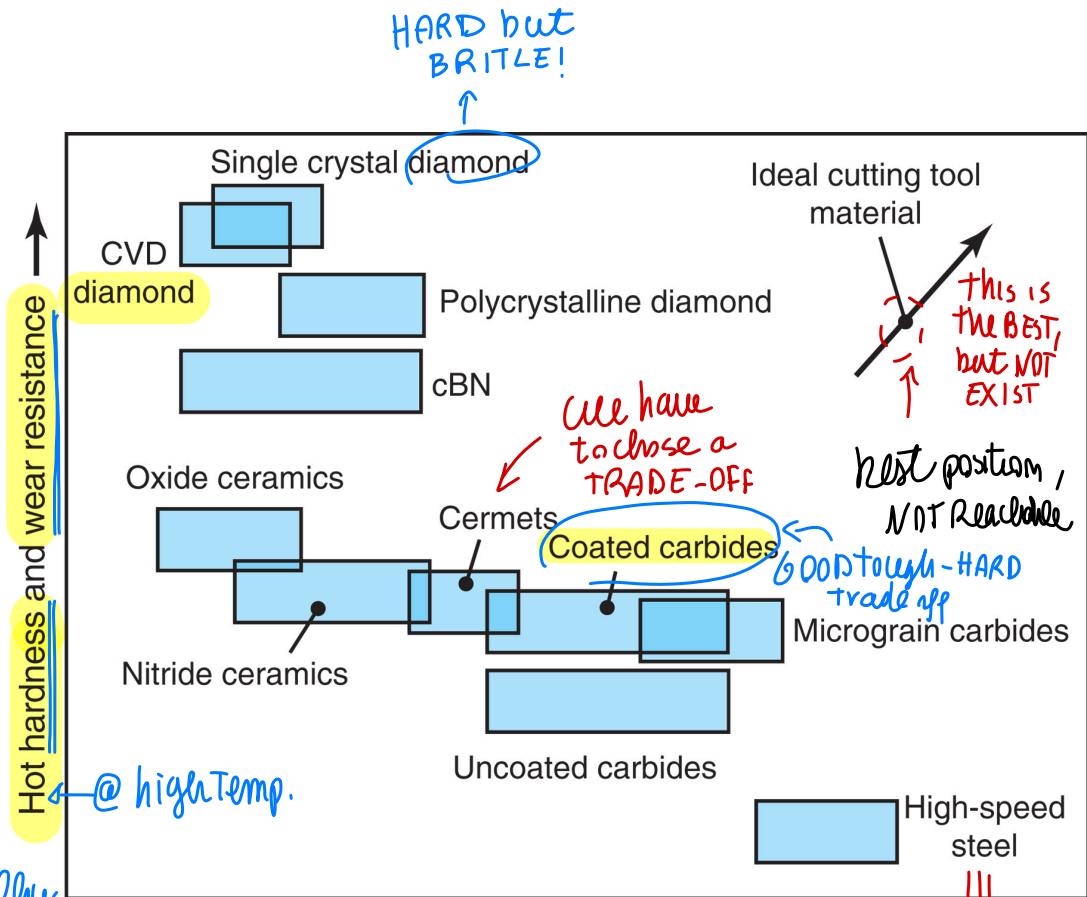
↓ behaviour of the material **HARDNESS**



First material used, in maching

new properties...

celllys that increase HRA with Temp. T°



Very tough but poor hardness respect Temperature



CUTTING TOOL MATERIALS

↓ choose material according to our situation

Less wear leads to:

- Increase of the production capacity
 - by increasing the removed chip volume,
 - by decreasing idle times for changing the worn tool. (less tool changes → time less)

- Decrease costs (✓ part)

- by decreasing the tool cost per part.

Decrease cost per unit of part → but you can have an
high cost tool

1 max MRR,
less change of tool
(NOT time lost)

↙ usable for many parts,
more productive,
so production cost reduced



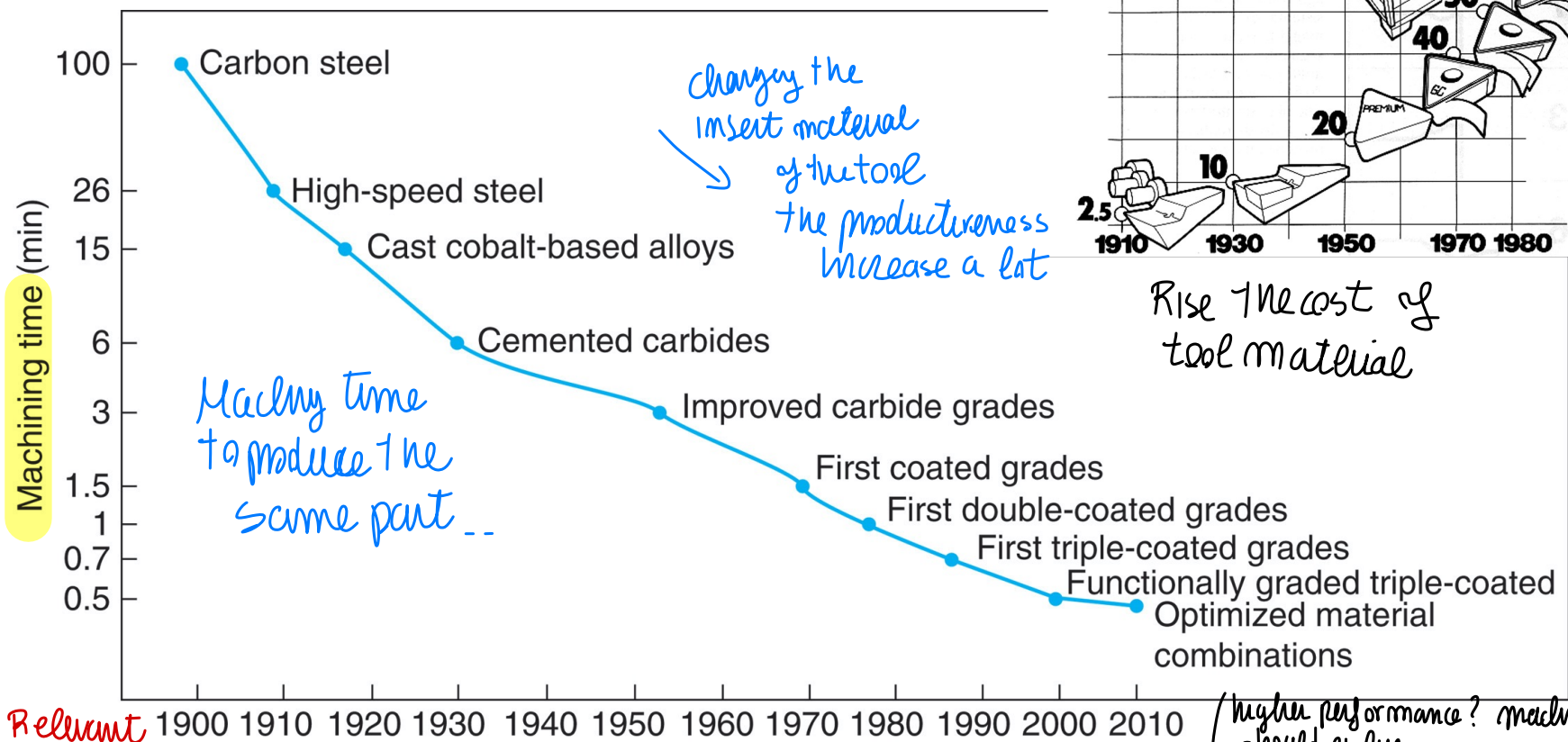
CUTTING TOOL MATERIALS

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different kind of material for machining over the year

Historical Evolution

how machining time decrease in history!



Relevant

ISSUE: evolution of tool material $\Rightarrow$ machine tool slow evolution
(De-evolution due to tool evolution of machine)

Machine tool evolution related to tool evolution



MACHINING OPTIMIZATION

We know the tool

life → so we can make OPTIMIZATIONS consideration

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select the objective function
Key Performance Indicators (KPI)
(objective function)

↑ depending on strategy

Best cutting condition that we can choose for our tools

System

- decisional variables
- constants
- relationships
- constraints

• tool life

Optimization → **Cutting Conditions**

↑ best cutting condition to choose

Approach:

- analytical
- numerical
- experimental

OUR POV.

→ in a simple situation!
we approach analytical the problem

⇒ IN OUR SITUATION,
TURNING OPERATION

TURNING

Decisional variables

V_c, f, a_p (cutting condition)

SYSTEM description

to choose for
max/min a
certain objective

Relationship

$$P_c = \frac{K_c A_D}{F_c} V_c$$

we need expressions

$$V_c T^m = C$$

$$P_c \leq \eta P_e$$

Power requirements

$$a_{pmin} \leq a_p \leq a_{pmax}$$

respect length of cutting edge... we use a portion
depth of cut limit, compatible with cutting tool

$$F_c = K_c A_D$$

$$f_{min} \leq f \leq f_{max}$$

feed limit, constrained by the tool manufacturer...

$$K_c = \frac{K_{cs}}{(h_D)^x}$$

$(h_D)^x$ in TURNING

$$h_D = f \sin(K_r)$$

tool failure limit V_{Blimit} (when change the tool)

(when defining a wear limit criterion) definition of

(OPTIMIZATION) (in MANUFACTURING production) TOOL LIFE

POSSIBLE objective functions

in that system we have to choose the value of our decisional variables

related to the tool change time

to max/min an obj. function
III OPTIMIZATION

cost C_p

production time T_p (for a single machining operation!)

mainly 3 elements
fixed time
cutting time

example: ||

Set-up of machines (put right tool on the TOCKET)

GENERAL FORM

prepare fixturing, hold the part -- prepare machine

$$T_p = T_h + T_c + \frac{T_t}{m_p}$$

we divide the total change time T_t for the part it can produce
number of PARTS done by a single cutting edge, overall
time is the tool life (that tool can be used)

$$T_p = T_h + \frac{V}{MRR} + \frac{T_t}{T} \frac{V}{MRR}$$

tool changing time

$$T_p = T_h + \frac{V}{MRR} \left(1 + \frac{T_t}{T} \right)$$

$$m_p = \frac{T}{T_c} = \frac{T}{V/MRR}$$

part done by a single tool

C_p

production cost

mainly 2 elements affecting the cost

machine usage cost
tool cost

(cost per cutting time to use that machine tool)

$$C_p = C_o \cdot T_p + \frac{C_T}{m_p} = C_o T_h + C_o \frac{V}{MRR} \left(1 + \frac{T_t}{T} \right) + \frac{C_T}{T} \frac{V}{MRR}$$

usage of that machine for T_{total} time

tool cost C_T

respect m_p parts made (spread the total tool cost or respect how many part I do)

$$C_p = \underbrace{C_0 T_h}_{\text{fixed cost (for FIXTIME)}} + \underbrace{\frac{C_0 V}{HRR} \left(1 + \frac{T_t}{T} + \frac{1}{T} \frac{C_t}{C_0} \right)}_{\text{cost} \times \text{number (adimensionum)}}$$

(cost per each part)

$C_0 = [\text{€}/\text{time}]$

$C_t = [\text{€}]$ (total cost)

$$C_p = C_0 T_h + \frac{C_0 V}{HRR} \left(1 + \frac{1}{T} \left(T_t + \frac{C_t}{C_0} \right) \right)$$

so the goal in OPTIMIZATION is to minimize the cost

$$F = \underbrace{\overline{F_0}}_{\text{FIXED value}} + \underbrace{2}_{\text{constant}} \underbrace{\frac{1}{HRR}}_{\text{constant}} \left(1 + \underbrace{\overline{Q}}_{\text{constant}} \right)$$

obj function form, shape of T_p, C_p

$\overline{T}$ total life (process parameters)

↑ with that general form... respect process parameter

HRR, T depends on process parameters

objective functions

OPTIMIZATION

$$\begin{cases} T_p = T_h + \frac{V}{HRR} \left(1 + \frac{T_t}{T} \right) \rightarrow \min \{T_p\} \\ C_p = C_0 T_h + \frac{C_0 V}{HRR} \left[1 + \frac{1}{T} \left(T_t + \frac{C_t}{C_0} \right) \right] \rightarrow \min \{C_p\} \end{cases}$$



OBJECTIVE FUNCTION

A) *Production Time* [T_p]

$$T_p = T_h + T_c + \frac{T_t}{n_p} = T_h + \frac{V}{MRR} + \frac{V \cdot T_t}{MRR \cdot T} = T_h + \frac{V}{MRR} \left(1 + \frac{T_t}{T} \right)$$

T : tool life [min.]

V : removed material volume [mm³]

$MRR = A_D \cdot v_c$: material removal rate [mm³ / min.]

T_h : fixed time [min.]

$T_c = \frac{V}{MRR}$: machining time [min.]

T_t : tool change time [min.]

$n_p = \frac{T}{T_c} = \frac{MRR \cdot T}{V}$: number of machined parts during tool life [min.]



B) *Production Cost* [C_p]

$$C_p = C_o T_p + \frac{C_t}{n_p} = C_o T_h + C_o \frac{V}{MRR} + C_o \frac{VT_t}{MRR \cdot T} + \frac{VC_t}{MRR \cdot T}$$

$$C_p = C_o T_h + C_o \frac{V}{MRR} \left[1 + \frac{1}{T} \left(T_t + \frac{C_t}{C_o} \right) \right]$$

C_o : production resource cost per minute (machine tool and human operator) [€ /min.]

C_t : average cost of the cutting edge [€ /cutting edge]



Max Power $F_c v_c \leq \eta \cdot P_e$

Max workpiece bending $b_{wp} \leq \varepsilon_{wp}$

Max tool bending $b_{tool} \leq \varepsilon_{tool}$

Feed range $f_{min} \leq f \leq f_{max}$

Cutting speed range $v_{c,min} \leq v_c \leq v_{c,max}$

Roughness requirement $\frac{f^2}{32r} \leq R_a$



- Taylor's Law $v_c T^n = C$
- Cutting Force $F_c = k_c A_D$
- ...



A) Production time [T_p]

B) Production cost [C_p]

$$\left. \begin{aligned} T_p &= T_h + \frac{V}{MRR} \left(1 + \frac{T_t}{T} \right) \\ C_p &= C_o T_h + C_o \frac{V}{MRR} \left[1 + \frac{1}{T} \left(T_t + \frac{C_t}{C_o} \right) \right] \end{aligned} \right\} \rightarrow F = F_0 + \frac{\alpha}{MRR} \left(1 + \frac{\varphi}{T} \right)$$

OBJECTIVE FUNCTION general shape

$$\bar{F} = F_0 + \frac{\alpha}{MRR} \left(1 + \frac{q}{T} \right)$$

respect each obj function

F	F_0	α	q
T_p	T_h	V volume to remove	T_t
C_p	$C_0 T_h$	$C_0 V$	$T_t + \frac{C_t}{C_0}$

We can minimize T_p, C_p both

$\Rightarrow$

to guarantee the OPTIMIZATION we have to MINIMIZE

Min F

$U_{c, a, p, f}$ find those parameters

Subj. to : $\left. \begin{array}{l} \text{---} \\ \text{---} \\ \text{---} \end{array} \right\}$ constraints

↑ minimize respect one objective function F

Min $T_p, \text{Min } C_p$

$U_{c, a, p, f}$

minimize both T_p, C_p together

Subj to $\left. \begin{array}{l} \text{---} \\ \text{---} \end{array} \right\}$ constraints

↑ multi-objective function optimization!

↓ to solve a multi-obj constrained problem there are different complex computational approaches...

different kind of approaches

We want

to OPTIMIZE CUTTING conditions for a single machine operation ↓

if consider a workpiece, you consider multiple machine operation to obtain the final part

↓ manufacturing system able to perform all operations...

set of machine tool respecting the requirements on the demand for production...

↑ scaling up the problem to complex situation...

→ choose the machine tools... to respect demand requirements!

↓ BIG PROBLEM



If the optimization parameter is the cutting speed:

$$\begin{array}{l} F = F_0 + \frac{\alpha}{MRR} \left(1 + \frac{\varphi}{T} \right) \\ T = T(v_c) \\ MRR = A_D v_c \end{array} \quad \Rightarrow \quad F = F_0 + \frac{\alpha}{A_D v_c} \left(1 + \frac{\varphi}{T(v_c)} \right)$$
$$F = F_0 + \frac{\alpha'}{v_c} \left(1 + \frac{\varphi}{T(v_c)} \right)$$

multiple machine tools respect the Requirements --

Remains in a **SIMPLE CASE**, solvable **analytically**...

Analytically?

Consider as variable V_c relevant variables define

unconstrained optimiz.

- (1 single variable V_c)
- (unconstrained optimiz. problem)

TURNING operation

$$MRR = A_D \cdot V_c \rightarrow \frac{\alpha}{A_D} = \alpha'$$

$$\Rightarrow \frac{\alpha}{MRR} = \frac{\alpha'}{V_c} \text{ to have explicitly } V_c$$

α' constant...

OBJECTIVE FUNCTION

$$F = F_0 + \underbrace{\left(\frac{\alpha}{MRR}\right)}_{\alpha'} \left(1 + \frac{\varphi}{T}\right)$$

$$F = F_0 + \frac{\alpha'}{V_c} \left(1 + \frac{\varphi}{T(V_c)}\right)$$

(derivative) ↓

OPTIMUM

$$\left\{ \begin{array}{l} \frac{dF}{dV_c} = 0 \rightarrow -\frac{\alpha'}{V_c^2} \left(1 + \frac{\varphi}{T}\right) - \frac{\alpha'}{V_c} \left(\frac{\varphi}{T^2} \frac{dT}{dV_c}\right) = 0 \\ \frac{d^2F}{dV_c^2} > 0 \text{ (to be MIN)} \end{array} \right.$$

$V_c \neq 0$ for shine, or we are not coming!

$$-\frac{\alpha'}{V_c^2} \left(1 + \frac{\varphi}{T} + \frac{\varphi}{T} \left(\frac{V_c}{T} \frac{dT}{dV_c}\right)\right) = 0$$

$$\frac{V_c}{T} \frac{dT}{dV_c} = \frac{d \ln T}{d \ln V_c}$$

$$(*) \quad 1 + \frac{\varphi}{T} \left(1 + \frac{d \ln T}{d \ln V_c}\right) = 0$$

to discuss this equation, by

using **Taylor's law**

$$V_c T^m = C \quad \text{as } \ln \text{ equation}$$

$$\ln V_c + m \ln T = \ln C$$

$$\ln T = -\frac{1}{m} \ln V_c + \frac{1}{m} \ln C$$

$$\frac{d \ln T}{d \ln V_c} = -\frac{1}{m}$$

it is a constant!

so the equation became

$$\Rightarrow (*) \quad 1 + \frac{\varphi}{T} \left(1 - \frac{1}{m}\right) = 0 \rightarrow 1 = \frac{\varphi}{T} \left(\frac{1}{m} - 1\right)$$

↓
 $\in [0,1]$

$m \in [0,1]$ by definition

general
OPTIMIZATION
formula
of V_{cOPT}

$$T_{OPT} = \varphi \left(\frac{1}{m} - 1\right)$$

↓ Taylor's law

$$V_{cOPT} = C / T_{OPT}^m$$

$$T_{opt}(T_p) = T_t \left(\frac{1}{m} - 1 \right) < T_{opt}(C_p) = \left(T_t + \frac{C_t}{C_o} \right) \left(\frac{1}{m} - 1 \right)$$

↑ respect $\min(T_p)$ > 0 $\leftarrow \min(C_p)$ $> T_t$ > 0

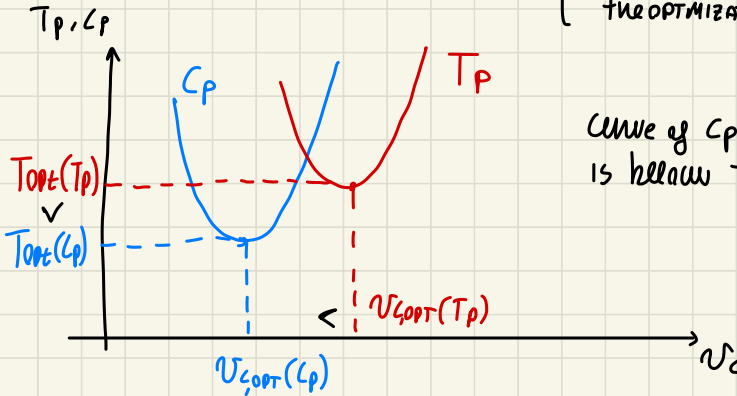
DISCUSSING the RESULT... $\leftarrow$ using the specific formula respect C_p, T_p

$$V_{COPT} = \frac{C}{T_{OPT}^m} \rightarrow \begin{cases} T_{OPT}(T_p) < T_{OPT}(C_p) \\ V_{COPT}(T_p) > V_{COPT}(C_p) \end{cases}$$

IF OPTIMIZING the cost we have a greater OPTIMAL tool life respect the OPTIMIZATION of the production T

IF PLOTTING the RESULTS

$T_p(V_c)$
 $C_p(V_c)$



other consideration about objective function

(F) MIN objective

$$\frac{dF}{dV_c} = Q$$

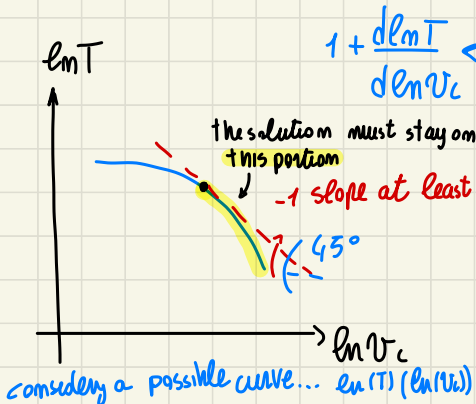
$$\frac{d^2F}{dV_c^2} > 0 \rightarrow \frac{d^2 \ln T}{d(\ln V_c)^2} < \frac{d \ln T}{d \ln V_c} \left(1 + \frac{d \ln T}{d \ln V_c} \right) = Q$$

< 0 to be Q that sum > 0 < 0 MUST BE NEG.

(we can introduce other possible obj functions)

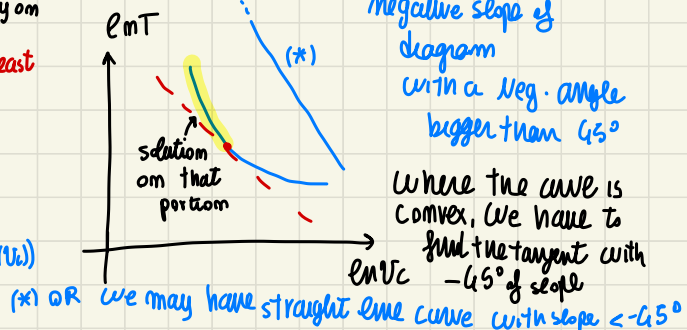
considering the II derivative, we improve the following constraints

equation valid despite Taylor's Law



$$1 + \frac{d \ln T}{d \ln V_c} < Q \Rightarrow \frac{d \ln T}{d \ln V_c} < -1$$

MUST BE a negative slope of diagram with a neg. angle bigger than 45°



IF this is valid...



$$\frac{\frac{d^2 \ell m T}{d(\ell m V_c)^2}}{\frac{d \ell m T}{d \ell m V_c}} \left(1 + \frac{d \ell m T}{d \ell m V_c} \right) > 0$$

$$\frac{d^2 \ell m T}{d(\ell m V_c)^2} < \frac{d \ell m T}{d \ell m V_c} \left(1 + \frac{d \ell m T}{d \ell m V_c} \right) \Rightarrow \frac{d^2 \ell m T}{d(\ell m V_c)^2} < \delta$$

with $\delta > 0$

so the solution will NOT depend on the curvature...

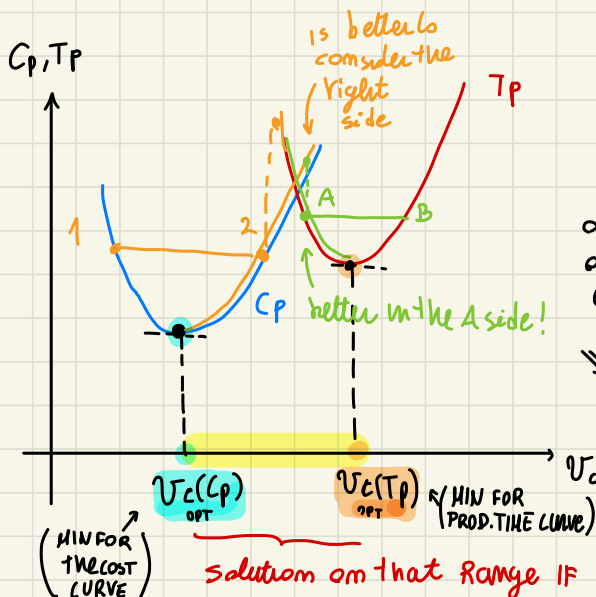
the curvature can be positive / Neg (related to $\delta > 0$) → JUST lower than δ
 or it can be a 0 curvature → straight line (STILL SATISFY δ)
 which satisfy the condition

general condition

satisfied by default from the Taylor Law ← it is a straight line, with a curvature, which satisfies $\frac{d^2 \ell m T}{d(\ell m V_c)^2} < \delta$, with a neg curvature, (< -1)

① { We can use TAYLOR'S LAW because it satisfy the CONDITION ABOUT I, II DERIVATIVE to find the min }

② 2nd comment ...



{ IN OPTIMIZATION
 you can have
 MULTI OBJECTIVE
 function → so
 multi-obj optimiz }

OR we can summarize one obj function differently, TRADE OFF of C_p, T_p by a single OBJ FUNCT. unique

⇒ we consider as
 objective function
 the PROFIT RATE

$$P_R = \frac{R - C_p}{T_p}$$

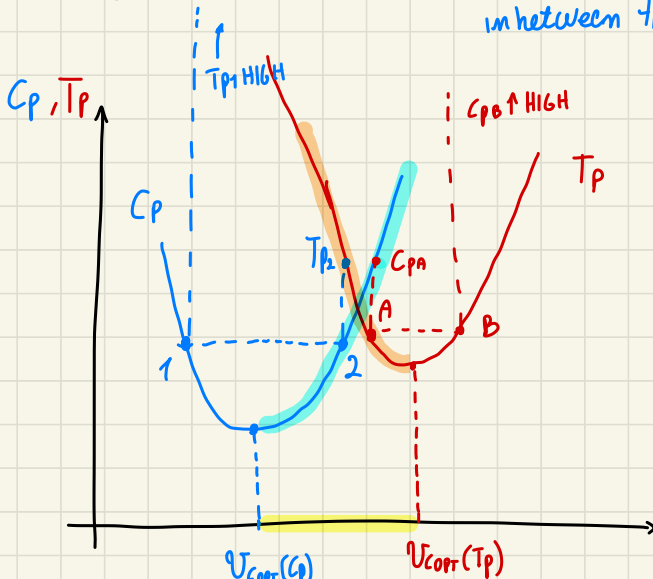
↓ Revenue

$P_R > 0$ obviously!

Both trying to do OPTIMIZATION of $\begin{cases} \text{a single objective function} \\ \text{MULTI-OBJ OPTIMIZ.} \end{cases}$



the found solution will be $V_{\text{OPT}} \in [V_{\text{OPT}}(C_p), V_{\text{OPT}}(T_p)]$
in between the two



• considering a couple situation
with same cost, we
prefer 1, 2?
same $C_{p1} = C_{p2}$
BUT respect T_p -- $T_{p1} \gg T_{p2}$
↓ so it is better to
consider the right portion
of the CURVE, $V_c > V_{\text{OPT}}(C_p)$

• reasoning go same Time,
 $T_{pA} = T_{pB} \rightarrow$
 $C_{pB} \gg C_{pA} \rightarrow$ is better A
so we prefer left side
 $V_c < V_{\text{OPT}}(T_p)$

so combining the
two condition
IT IS
BETTER to
stay here

we can PROVE qualitatively that: define the **PROFIT RATE** obj. function

obviously we
WANT $P_R > 0$ to have
Revenue
on company...
↓
 $P_R < 0$ IT BAD!

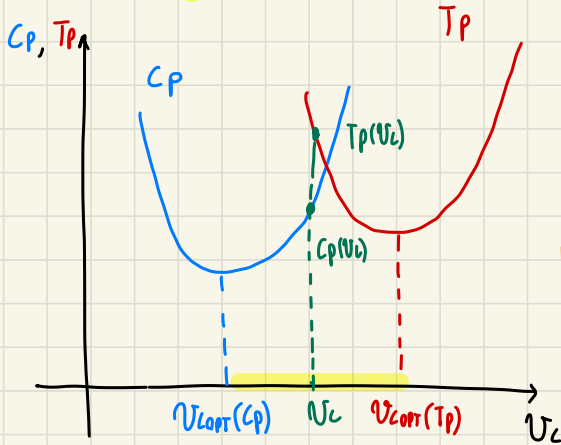
$$\boxed{P_R = \frac{R - C_p}{T_p}}$$

selling
revenue

TRADE-OFF
between cost
and TIME

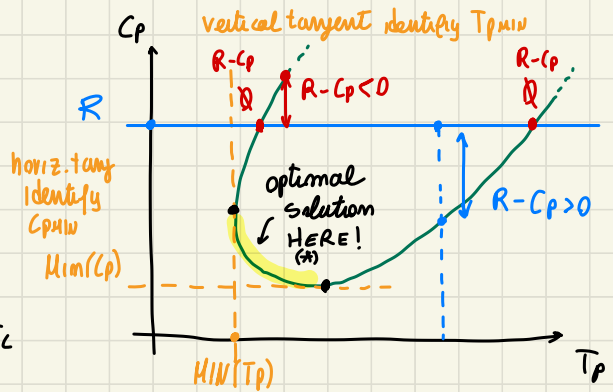
What we try to MAXIMIZE
finding a TRADE OFF

to discuss P_a OPTIMIZATION



→ from the two curves...

fixing V_c we have a certain C_p, T_p Point on a Graph $C_p(T_p)$



Double value function,

{ because of C_p fixed → 2 corresponding T_p
 and T_p fixed → 2 C_p corresponding }

← and considering the different V_c , we obtain the following function

↓
 saying that $V_{c, \text{opt}} \in [V_{c, \text{opt}}(C_p), V_{c, \text{opt}}(T_p)]$ → between the two MIN of T_p, C_p
 OPTIMAL TRADE OFF

WE CAN PROVE THAT THE OPTIMUM IS HERE (x)

CONSIDERING THE REVENUE (of same unit of measure of cost (C_p))

to compare with C_p → R independent respect process parameters → IT DEPENDS ON MARKET

$R = \text{constant}$
 (INDEPENDENT from T_p or $V_c \dots$) always the same

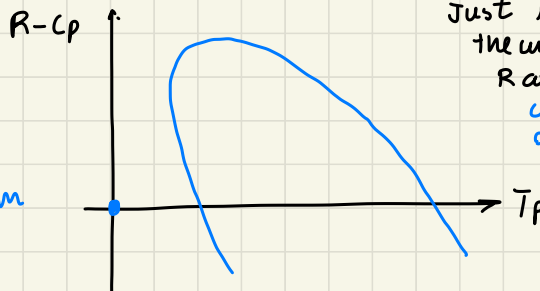
so, by considering the difference $R - C_p(T_p)$

{ $R - C_p > 0$ when the curve is below Revenue R → positive x o b T function
 $R - C_p < 0$ if above revenue }

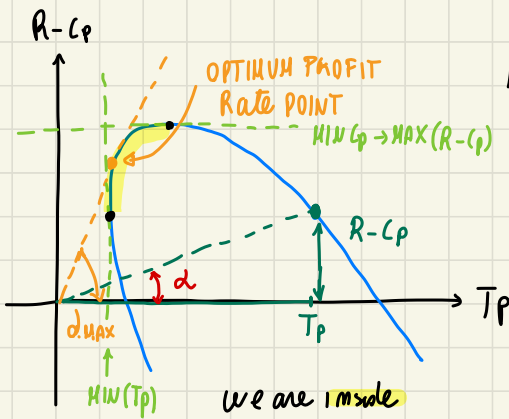
↓
 we can
 PLOT $(R - C_p)$ respect T_p

IT IS LIKE THE SAME CURVE...

MIRROR the function respect an axis on the Revenue line



Just MIRROR
 the curve respect
 R axis...
 with position of
 origin corresponding
 to R line
 ↓
 when $C_p = R$
 $R - C_p = 0$ ORIGIN



we are inside
the OPTIMAL Region
of the curve

considering a generic point on the curve
IF I'm interested in

$$\left\{ P_R = \frac{R - C_p}{T_p} \right\} \text{ OBJ function}$$

$$P_R = \tan(\alpha)$$

and the optimum is where
 $\tan(\alpha) \rightarrow \text{MAX}$, so
where d is MAX

↑ TRADE-OFF between
 C_p, T_p we find that
OPTIMAL condition in
between the two OPTIMAL
condition!

$$\left. \begin{aligned} V_C &\geq V_{C_{\text{OPT}}}(C_p) \\ V_C &\leq V_{C_{\text{OPT}}}(T_p) \end{aligned} \right\} \rightarrow$$

better interesting Region where

to stay to OPTIMIZE our PROCESS! $\rightarrow$ to solve we use KPI
coherent to our goal of production

We can define a lot of others objectives functions

- SUSTAINABILITY $\rightarrow$ Energy consumption $\Rightarrow \text{minim}\{E_p\}$

Built the obj function $\rightarrow$ perform OPTIMIZATION,

We discuss the simplest case BASIC

- UNCONSTRAINED
- with ONE decisional variable

in general

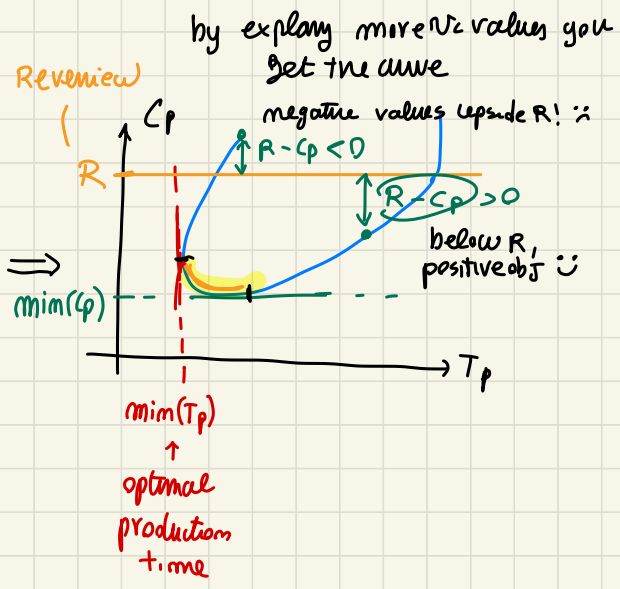
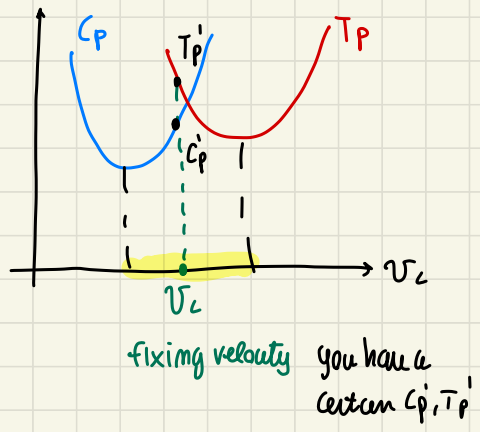
- **CONSTRAINED PROBLEM**
- Multiple decisional var.
- Multiple obj function

↑

MULTI-OBJ OPTIMIZATION
IN A CONSTRAINED
PROBLEM

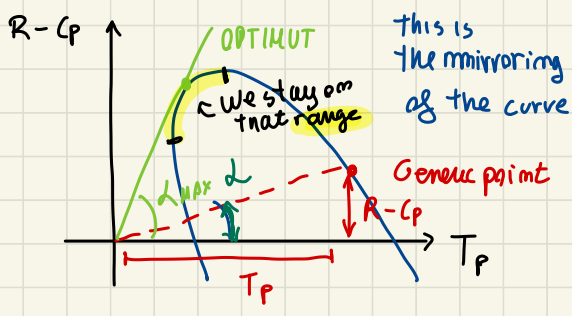
COMPLEX!

-to discuss PR optimization:



$R \approx \text{constant, independent}$

we can plot



this is the mirroring of the curve $C_p(T_p)$

I'm interested in $\frac{R - C_p}{T_p} = \tan \alpha$

optimum is when α is max!

another possible object can be to minimize energy consumption



UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

$$\frac{dF}{dv_c} = 0$$

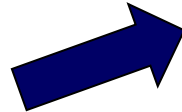
$$\frac{d}{dv_c} \left[F_0 + \frac{\alpha'}{v_c} \left(1 + \frac{\varphi}{T(v_c)} \right) \right] = 0$$

$$-\frac{\alpha'}{v_c^2} \left(1 + \frac{\varphi}{T} \right) - \frac{\alpha' \varphi}{v_c T^2} \frac{dT}{dv_c} = 0$$

$$-\frac{\alpha'}{v_c^2} \left(1 + \frac{\varphi}{T} + \frac{\varphi v_c}{T^2} \frac{dT}{dv_c} \right) = 0$$

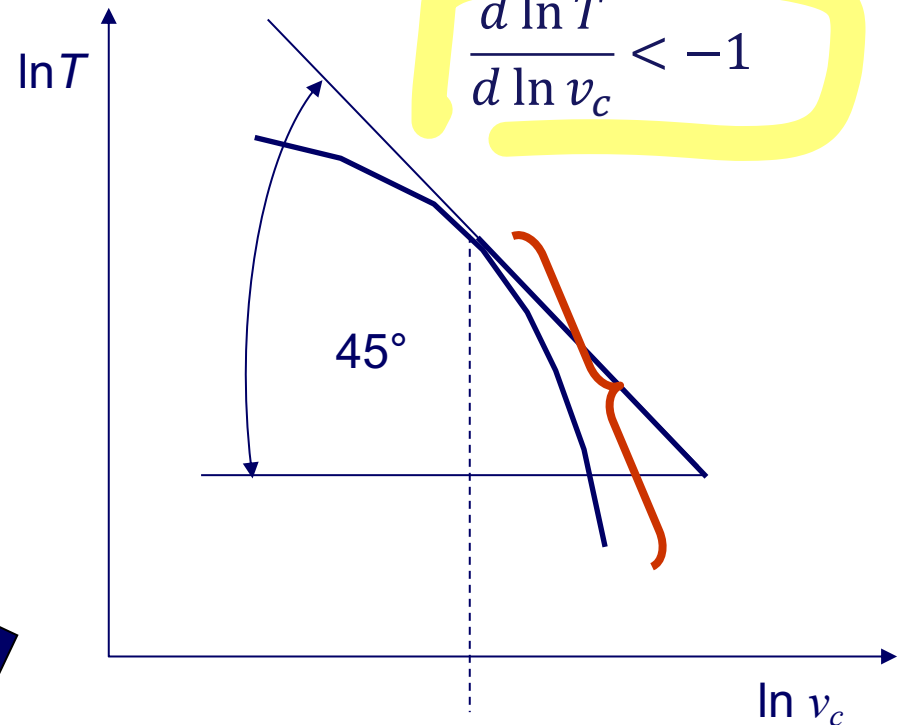
$$1 + \frac{\varphi}{T} + \frac{\varphi v_c}{T} \frac{dT}{dv_c} = 0$$

$$1 + \frac{\varphi}{T} \left(1 + \frac{d \ln T}{d \ln v_c} \right) = 0$$



An extreme point exists if:

$$\frac{d \ln T}{d \ln v_c} < -1$$





UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

Condition for the minimum

$$\frac{d^2 F}{dv_c^2} > 0$$

since $\frac{d \ln T}{d \ln v_c} < -1$

$$\frac{d^2(\ln T)}{d(\ln v_c)^2} < \frac{d(\ln T)}{d(\ln v_c)} \left(1 + \frac{d(\ln T)}{d(\ln v_c)} \right)$$

then $\frac{d^2(\ln T)}{d(\ln v_c)^2} < \delta$

< -1 < -1

< 0

With $\delta > 0$



UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

Let's consider the Taylor's law: $T^n(v_c) = \frac{C}{v_c}$

In this case the value $\frac{d \ln T}{d \ln v_c}$ is: $n \ln T = \ln C - \ln v_c$
 $\ln T = \frac{1}{n} \ln C - \frac{1}{n} \ln v_c$
 $\frac{d \ln T}{d \ln v_c} = -\frac{1}{n}$

It follows that: $1 + \frac{\varphi}{T} \left(1 - \frac{1}{n}\right) = 0$

$$T_{opt} = \varphi \left(\frac{1}{n} - 1 \right) \rightarrow v_{c,opt} = \frac{C}{T_{opt}^n}$$



UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

The optimal condition is structural, thus it holds both for T_p and C_p .

Since:

$$\varphi(T_p) = Tt < \varphi(C_p) = Tt + C_t/C_0$$

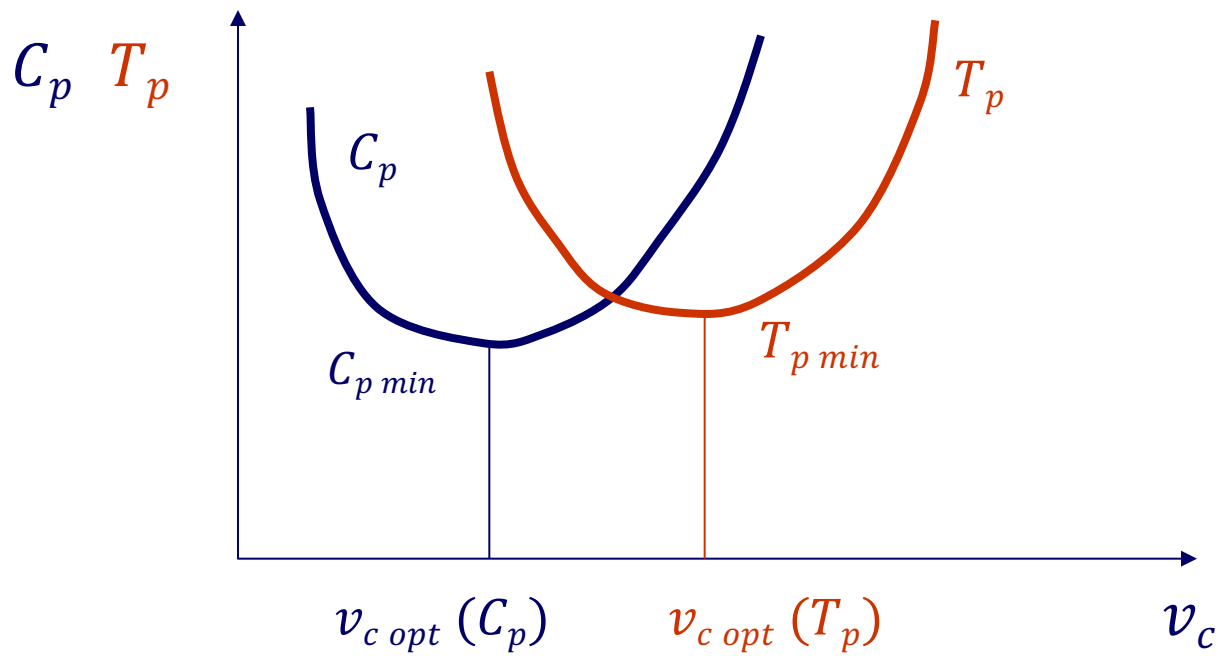
Then, it follows that:

$$T_{opt}(T_p) < T_{opt}(C_p)$$

$$v_{c\ opt}(T_p) > v_{c\ opt}(C_p)$$

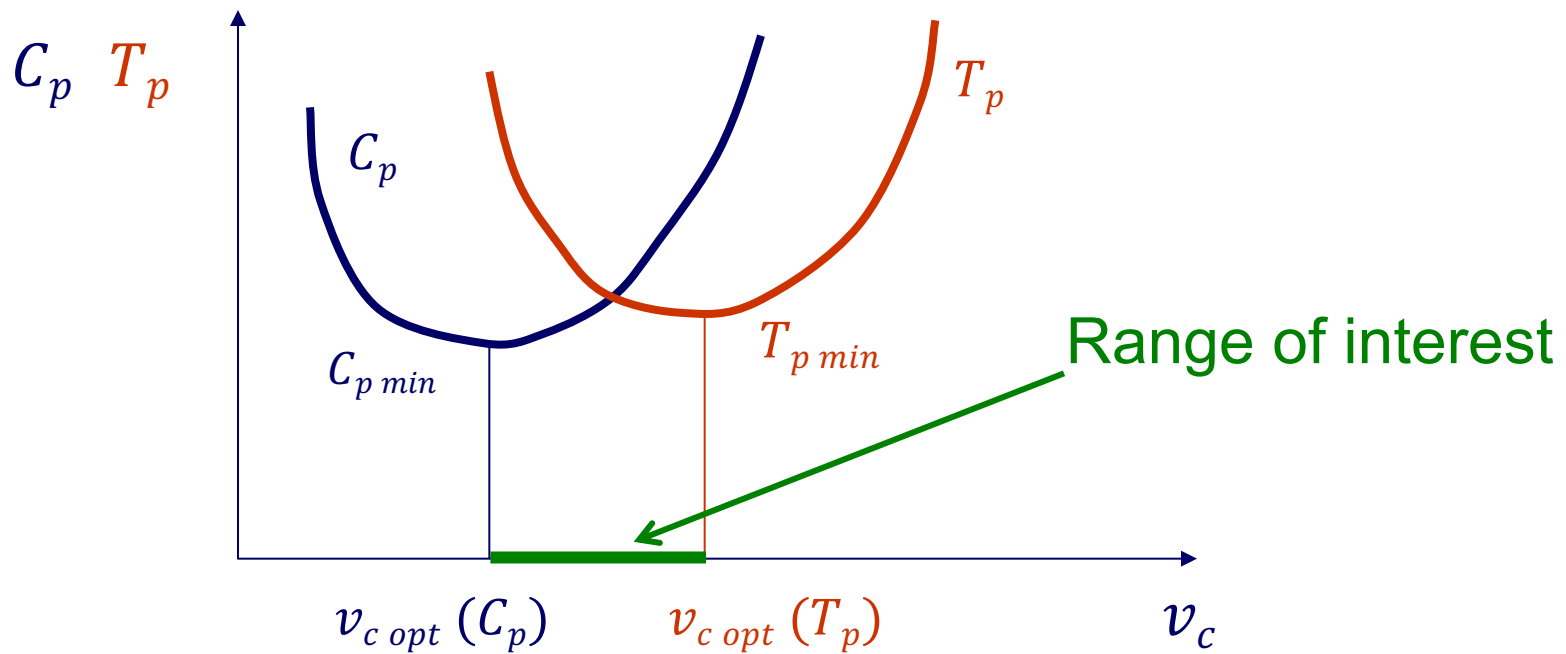


UNCONSTRAINED OPTIMIZATION: 1! VARIABLE





UNCONSTRAINED OPTIMIZATION: 1! VARIABLE





C) **Profit Rate** [P_r]

$$P_r = \frac{R - C_p}{T_p}$$

where:

R = revenue

$$T_p = T_h + \frac{V}{MRR} \left(1 + \frac{T_t}{T} \right)$$

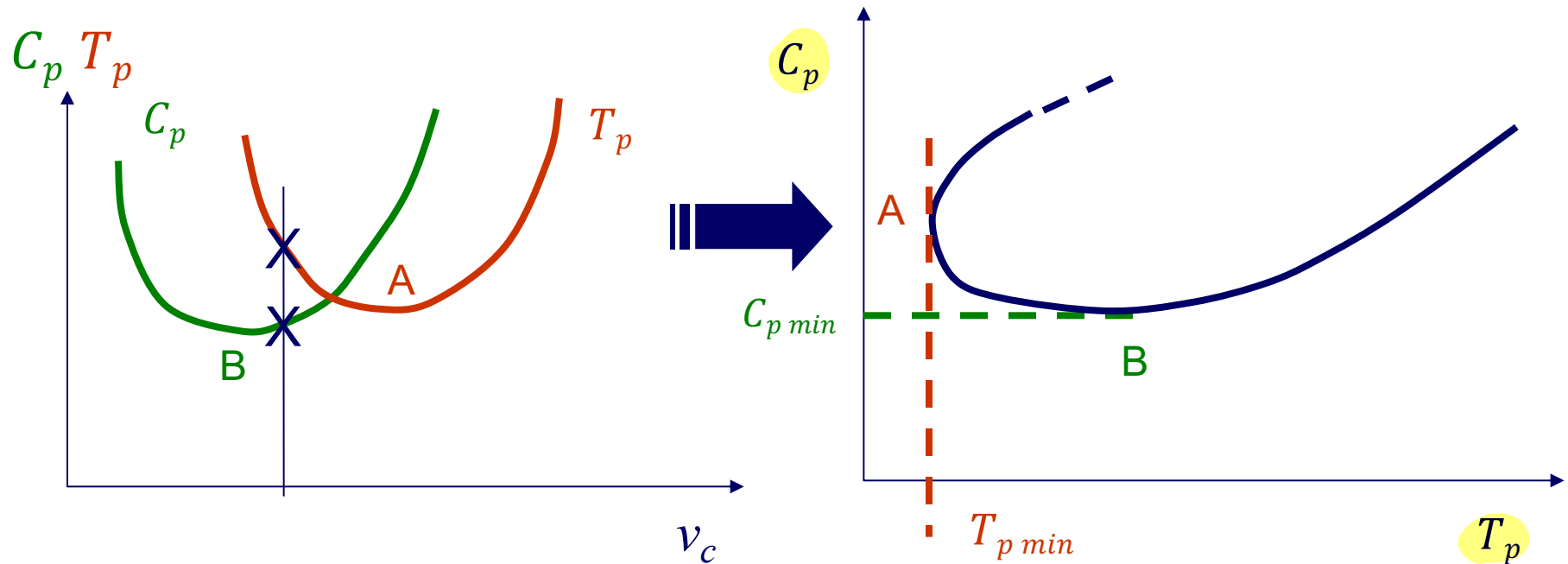
$$C_p = C_o T_h + C_o \frac{V}{MRR} \left[1 + \frac{1}{T} \left(T_t + \frac{C_t}{C_o} \right) \right]$$



UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

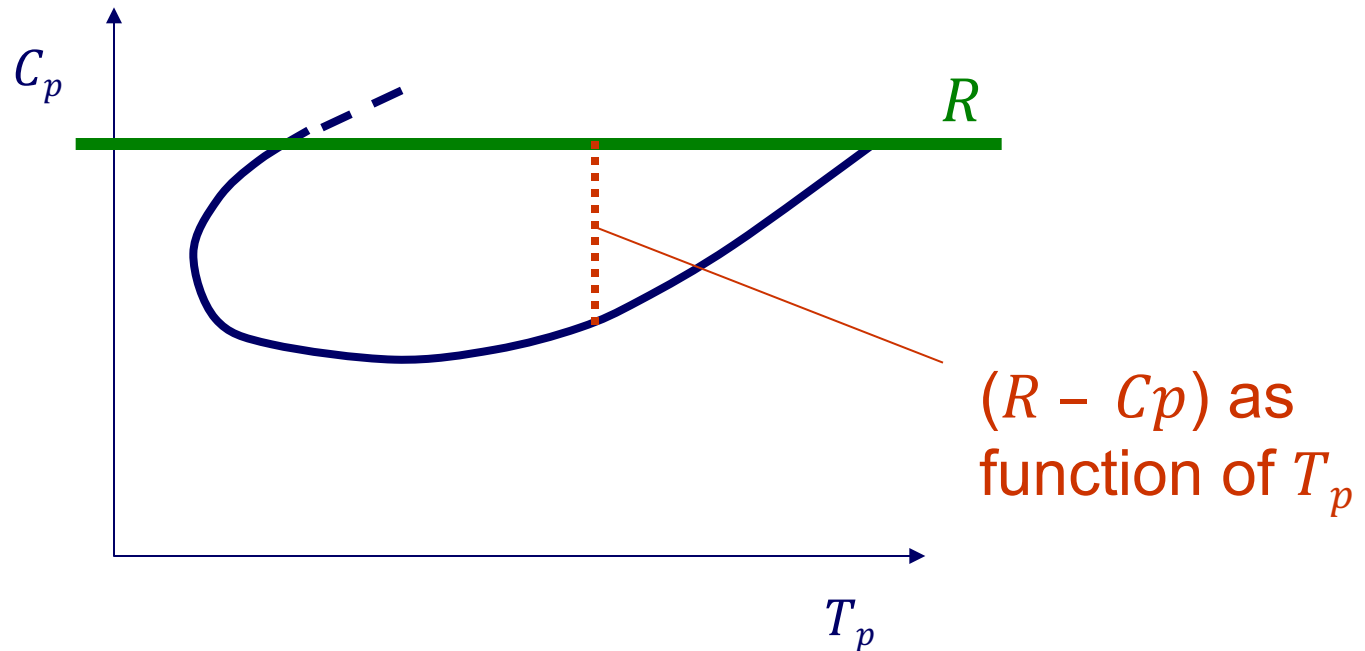
C) Profit Rate [P_r]

$$P_r = \frac{R - C_p}{T_p}$$



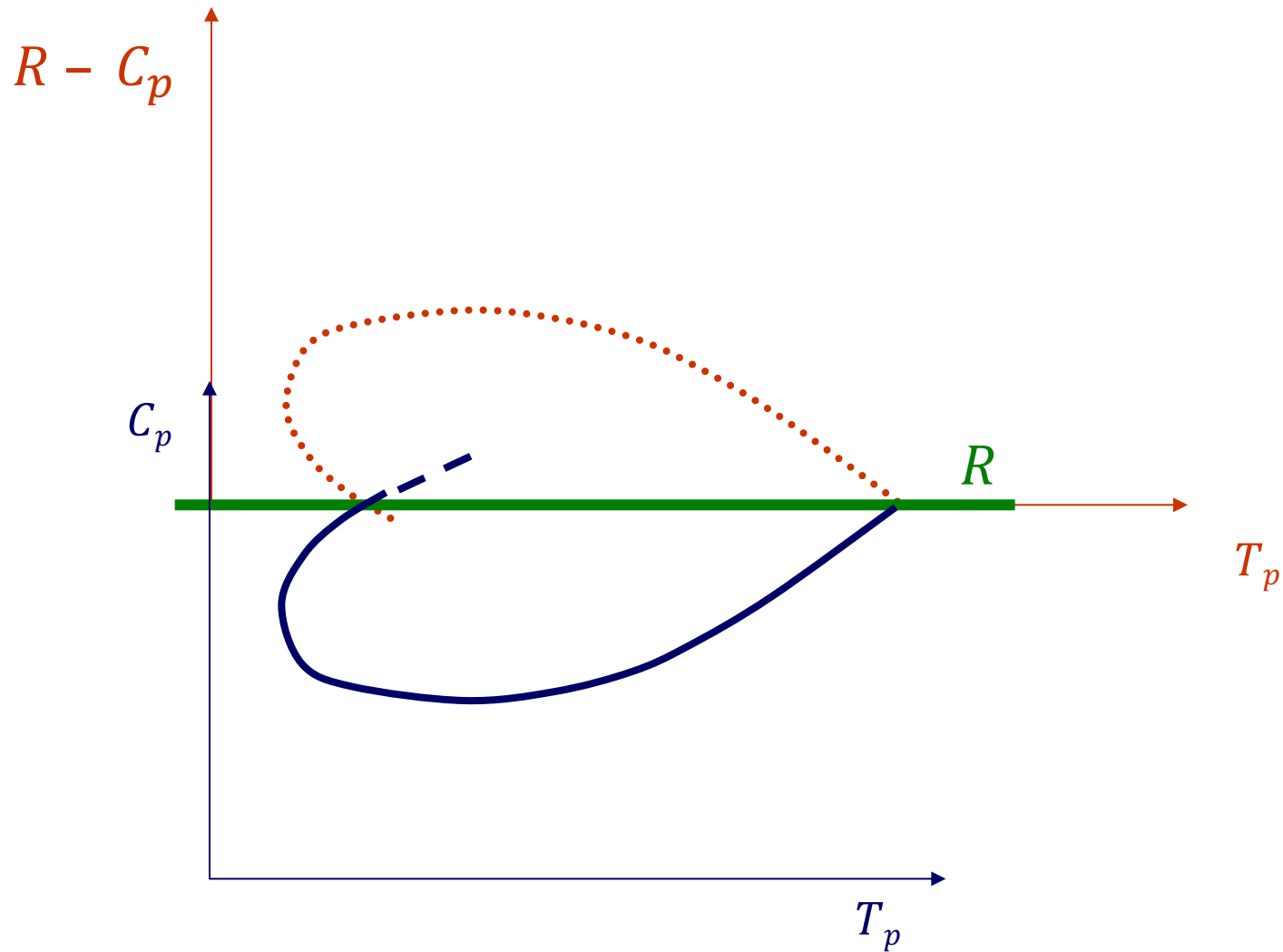


Assuming constant revenue R



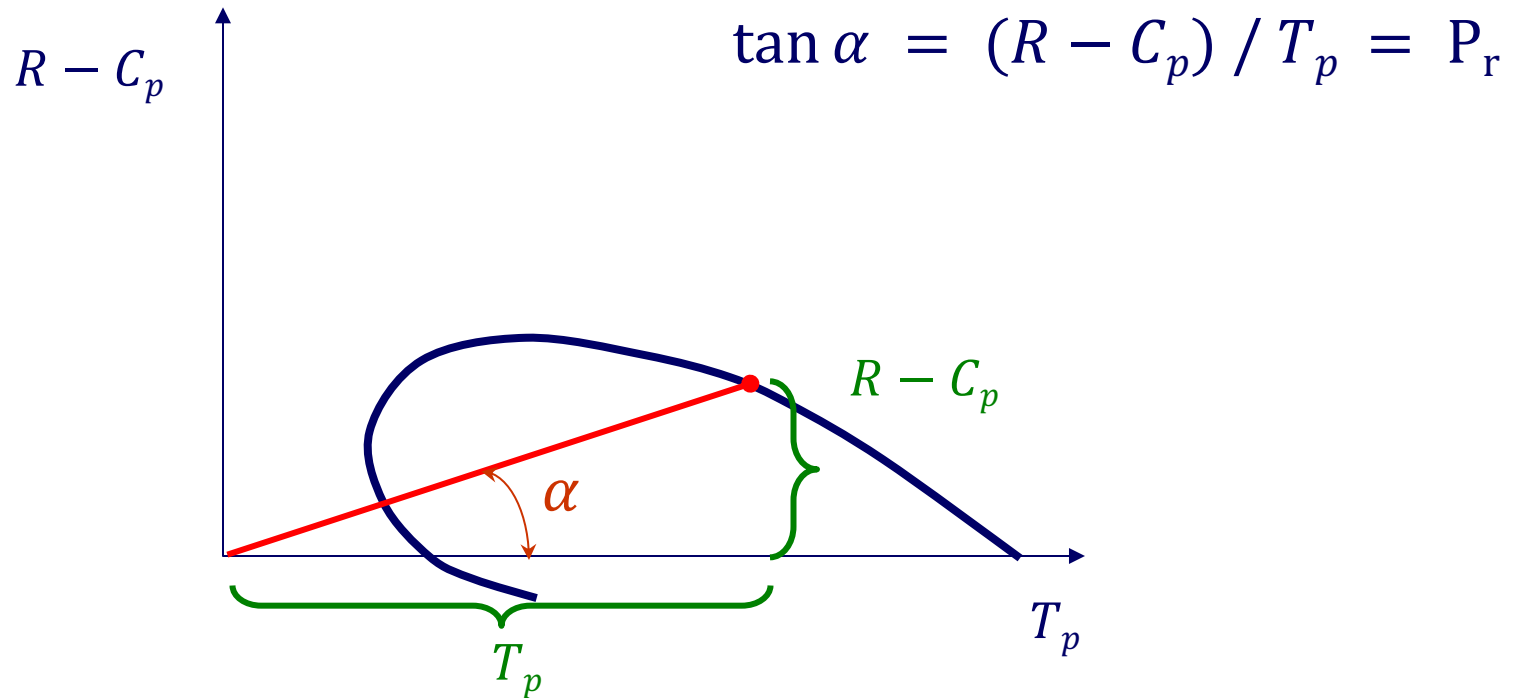


UNCONSTRAINED OPTIMIZATION: 1! VARIABLE



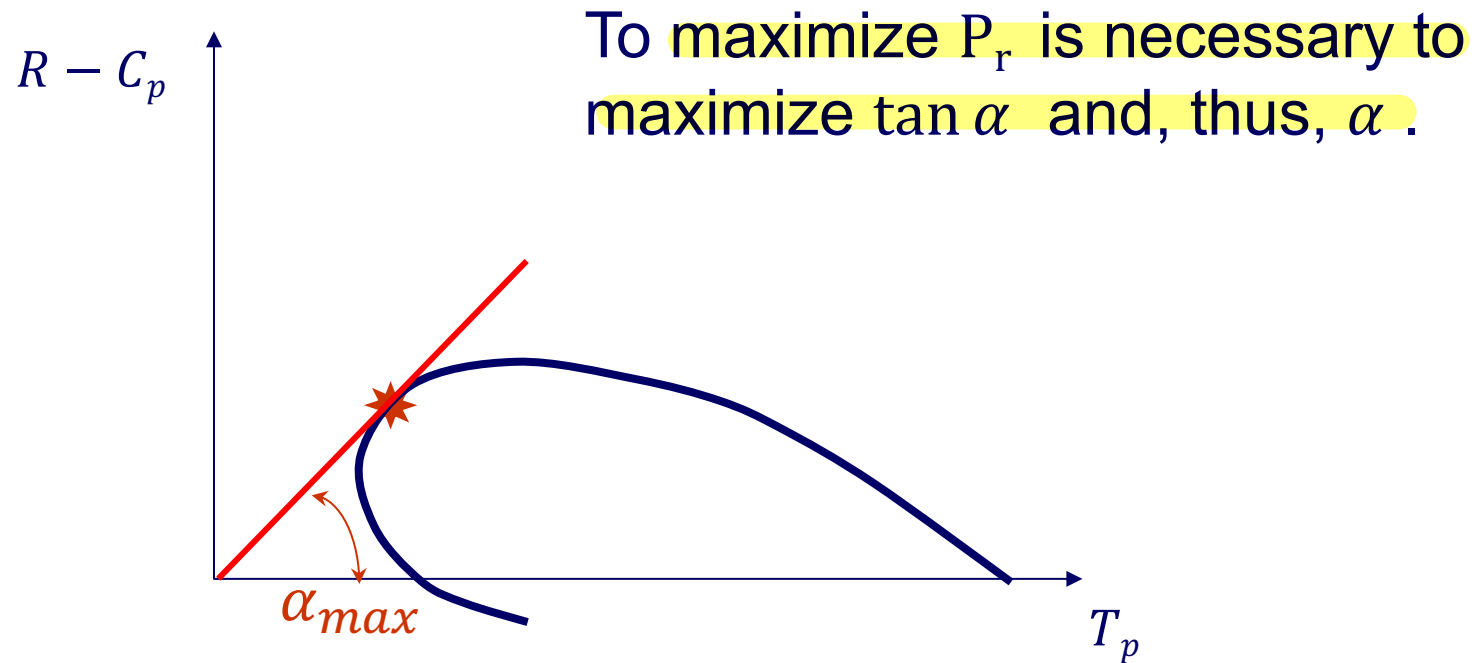


UNCONSTRAINED OPTIMIZATION: 1! VARIABLE





UNCONSTRAINED OPTIMIZATION: 1! VARIABLE





UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

The P_r optimal condition is between the ones that minimize C_p and T_p (range of interest).

